

ANSWERS & EXPLANATIONS
GENERAL STUDIES (P) TEST – 6335

- Q 1. B**
- The Pabna Revolt of 1873 took place in Bengal's Pabna district, now in Bangladesh. Under the British Permanent Settlement of 1793, zamindars became landowners, while peasants had to pay rent. This land revenue system led to heavy exploitation, as zamindars increased rents, imposed illegal taxes (abwabs), and forced peasants into labour (begar).
 - The peasants, already suffering from economic hardships and frequent famines, eventually rose in revolt against these oppressive practices. One of the key leaders, Ishan Chandra Roy (Ishan Raja), along with followers like Koodi Molla and Shambhu Nath Pal, declared their parganas independent from zamindari control and envisioned establishing a local government. **Hence, statement 1 is not correct.**
 - The Pabna Revolt gradually came to an end in the mid-1870s as the British colonial administration, recognizing the widespread nature of the unrest, began to address the grievances of the peasants. The revolt's decline was marked by a combination of government interventions, including military suppression and legal reforms. **Hence, statement 2 is not correct.**
 - The enactment of the Bengal Tenancy Act in 1885, which provided some protection to peasants against arbitrary evictions and illegal cesses, was a significant outcome influenced by the revolt. **Hence, statement 3 is correct.**
- Q 2. B**
- As per the ancient treatises, dance is considered as having three aspects: *natya*, *nritya* and *nritta*. ***Natya* highlights the dramatic element** and most dance forms do not give emphasis to this aspect today with the exception of dance-drama forms like Kathakali. ***Nritya* is essentially expressional, performed specifically to convey the meaning of a theme or idea.** *Nritta* on the other hand, **is pure dance where body movements do not express any mood (bhava), nor do they convey any meaning.** To present *nritya* and *natya* effectively, a dancer should be trained to communicate the *navarasas*. These are: love (*shringaara*), mirth (*haasya*), compassion (*karuna*), valour (*veera*), anger (*raudra*), fear (*bhayanak*), disgust (*bibhatsa*), wonder (*adbhuta*) and peace (*shaanta*). **Hence, only statement 3 is correct.**
- Q 3. C**
- The Department for Promotion of Industry and Internal Trade (DPIIT) is a central government department under the Ministry of Commerce and Industry in India. It is responsible for formulation and implementation of promotional and developmental measures for growth of the industrial sector, keeping in view the national priorities and socio-economic objectives. **Hence, statement 1 is correct.**
 - While individual administrative ministries look after the production, distribution, development and planning aspects of specific industries allocated to them, DPIIT is responsible for the overall industrial policy. It is also responsible for facilitating and increasing the foreign direct investment (FDI) flows to the country. **Hence, statement 2 is correct.**

- The department in its current form came into being on 27 January 2019, when the erstwhile Department of Industrial Policy & Promotion was renamed to Department for Promotion of Industry and Internal Trade (DPIIT) after internal trade was added to its mandate.
- The department was originally established in 1995, and was reconstituted in the year 2000 with the merger of Department of Industrial Development.
- DPIIT is the nodal Department in Government of India for coordinating and implementing programmes with the United Nations Industrial Development Organization (UNIDO) in India. The department also coordinates with apex Industry Associations such as Federation of Indian Chambers of Commerce and Industry, Confederation of Indian Industry, ASSOCHAM in their activities relating to promotion of industrial cooperation and to stimulate FDI into India.
- DPIIT is also responsible for intellectual property rights relating to patents, designs, trademarks, copyrights, layout-designs of integrated circuits, and geographical indication of goods, and oversees the initiative relating to their promotion and protection.

- Q 4. C**
- Biosphere reserves are areas identified by national governments for conserving biodiversity and promoting sustainable development. They have been described as ‘learning places for sustainable development’. They are sites for testing interdisciplinary approaches to understanding and managing changes and interactions between social and ecological systems, including conflict prevention and management of biodiversity. Biosphere reserves include terrestrial, marine and coastal ecosystems
 - The country has 18 Biosphere Reserves covering 91,425 sq. km, with 13 recognised by UNESCO.
 - India has 18 notified Biosphere Reserves covering approximately 91,425 sq. km, of which 13 are recognised by UNESCO’s World Network of Biosphere Reserves (WNBR).
 - List of India’s 13 UNESCO-Recognised Biosphere Reserves:
 - 1. Nilgiri (Tamil Nadu, Kerala, Karnataka)
 - 2. Gulf of Mannar (Tamil Nadu)
 - 3. Sundarbans (West Bengal)
 - 4. Nanda Devi (Uttarakhand)
 - 5. Nokrek (Meghalaya)
 - 6. Pachmarhi (Madhya Pradesh)
 - 7. Simlipal (Odisha)
 - 8. Achanakmar-Amarkantak (Madhya Pradesh, Chhattisgarh)
 - 9. Great Nicobar (Andaman & Nicobar Islands)
 - 10. Agasthyamalai (Kerala, Tamil Nadu)
 - 11. Khangchendzonga (Sikkim)
 - 12. Panna (Madhya Pradesh)
 - 13. Cold Desert (Himachal Pradesh)
 - Manas Biosphere Reserve is a nationally designated Biosphere Reserve and a UNESCO World Heritage Site; it has not been included in UNESCO’s World Network of Biosphere Reserves (WNBR) under the Man and the Biosphere (MAB) Programme. Hence, point 2 is not correct.
 - **Hence, option (c) is the correct answer.**

- Q 5. C** The Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization is a 2010 supplementary agreement to the Convention on Biological Diversity (CBD). It provides a transparent legal framework for implementing one of the three main objectives of the CBD: the fair and equitable sharing of benefits derived from the use of genetic resources.

The protocol establishes obligations for its contracting parties regarding:

- Access: Creating predictable conditions and clear procedures for prior informed consent (PIC).
- Benefit-sharing: Ensuring that benefits (monetary or non-monetary) are shared based on mutually agreed terms (MAT).
- Compliance: Supporting adherence to domestic legislation and monitoring the utilization of resources after they leave the providing country.

Hence, option (c) is the correct answer.

- Q 6. C** What is Greedflation?

- Greedflation refers to a situation where firms increase prices beyond what is justified by rising costs, primarily to expand profit margins. Unlike traditional inflation drivers, greedflation emphasises corporate pricing power rather than demand pressure or input cost escalation.
- **Hence, option (c) is the correct answer.**

How Greedflation Operates

In periods of economic stress or uncertainty—such as pandemics, wars, or supply-chain disruptions—companies often face higher costs. However, under greedflation:

- Firms use the general inflationary environment as cover
- Prices rise more than proportionately to actual cost increases
- Profit margins widen even when input costs stabilise or fall
- This behaviour is more common in concentrated markets where competition is limited and consumers have fewer alternatives.

How Greedflation Differs from Other Types of Inflation

- Demand-pull inflation arises when excess demand pushes prices up.
- Cost-push inflation results from higher input costs like wages, fuel, or raw materials.

Greedflation, in contrast, occurs when:

- Costs may rise modestly or temporarily
- Prices rise sharply and persistently
- Corporate profits increase faster than costs
- Thus, profit expansion—not cost pressure—is the defining feature.

- Q 7. C**
- The Press Council of India is a statutory quasi-judicial autonomous authority mandated by the Parliament for the twin objectives of preserving the freedom of the press and maintaining and improving the standards of newspapers and the news agencies in India exercising equal quasi-judicial functions over the authorities as well the press person. **Hence, statement 1 is correct.**
 - It comprises a Chairman and 28 members. While the Chairman has, by convention, be a sitting or retired judge of the Supreme Court of India, of the 28 members, 20 represent various segments of the

Press and eight overseeing the readers' interest, are representatives of the two Houses of Parliament (3 Lok Sabha and 2 Rajya Sabha) and premier literary and legal bodies of the country. **Hence, statement 2 is correct.**

- The Council has its own funds for the performance of its functions under the Act comprises the fee collected from newspapers, other receipts and also grants-in-aid from the central government.
- There is also an advisory role on the Council in that it can, either suo-motu or on a reference made to it by the government under Section 13(2) of the Act, undertake studies and express its opinion in regard to any bill, legislation, law or other matters concerning the press and convey its opinion to the government or the persons concerned. Also, in matters of public importance concerning its statutory responsibilities (preserving the freedom of the press and maintaining and improving the standards of newspapers and news agencies), the Council may suo-moto take an on-the-spot inquiry. **Hence, statement 3 is correct.**
- It is also empowered to make such observations as it may think fit in respect of the conduct of any authority, including the government, for interfering with the freedom of the press so prescribed under Section(4). The decisions of the Council are final and cannot be questioned in any court of law.

Q 8. B Schedule II of the Indian Constitution contains provisions related to the allowances, privileges, and emoluments of

- the **President**,
- Governors of States,
- Speaker and the Deputy Speaker of the House of the People and the
- **Chairman and the Deputy Chairman of the Council of States** and the
- Speaker and the Deputy Speaker of the Legislative Assembly and the
- Chairman and the Deputy Chairman of the Legislative Council of a State,
- **the Judges of the Supreme Court and the High Courts and**
- **the Comptroller and Auditor-General of India.**
- **Hence option (b) is the correct answer.**

Q 9. D • Echolocation is a technique used by bats, dolphins and other animals to determine the location of objects using reflected sound. This allows the animals to move around in pitch darkness, so they can navigate, hunt, identify friends and enemies, and avoid obstacles.

- Many animals use echolocation—a form of biological sonar—to navigate, forage, and communicate, especially in darkness or murky environments. By emitting high-frequency sounds and interpreting the returning echoes, they determine object location, size, and distance. Key users include bats, odontocetes (toothed whales/dolphins), shrews, and certain birds.

Key Animals Utilising Echolocation:

- Bats: Use ultrasonic clicks to fly, avoid obstacles, and catch small insects in complete darkness.
- Dolphins and Whales: Utilise clicks to navigate and locate prey in dark or muddy, deep ocean waters.
- Shrews and Tenrecs: Use lower-intensity sounds to navigate burrows and find food.
- Oilbirds and Swiftlets: Navigate dark caves using audible clicking sounds, making them unique bird species that use this technique.

- Pygmy Dormouse: Specifically, the Chinese pygmy dormouse (*Typhlomys cinereus*) and the Vietnamese pygmy dormouse, these nearly blind, tree-dwelling rodents use ultrasonic squeaks for arboreal navigation.
- Aye-Ayes: This nocturnal primate uses percussive foraging, tapping on trees with its specialised finger and listening to the echoes to locate hollow canals and grubs inside.
- **Hence, option (d) is the correct answer.**

Q 10. B ARMED FORCES AND FUNDAMENTAL RIGHTS

- **Article 33 empowers the Parliament to restrict or abrogate the fundamental rights of the members of armed forces, para-military forces, police forces, intelligence agencies and analogous forces.** The objective of this provision is to ensure the proper discharge of their duties and the maintenance of discipline among them. The **power to make laws under Article 33 is conferred only on Parliament and not on state legislatures. Hence statement 1 is not correct.**
- Any such law made by Parliament cannot be challenged in any court on the ground of contravention of any of the fundamental rights. **Hence statement 2 is correct.**
- Accordingly, the Parliament has enacted the Army Act (1950), the Navy Act (1950), the Air Force Act (1950), the Police Forces (Restriction of Rights) Act, 1966, the Border Security Force Act and so on. These impose restrictions on their freedom of speech, right to form associations, right to be members of trade unions or political associations, right to communicate with the press, right to attend public meetings or demonstrations, etc.
- The expression 'members of the armed forces' also covers such employees of the armed forces as barbers, carpenters, mechanics, cooks, chowkidars, bootmakers, tailors who are non-combatants.
- A parliamentary law enacted under Article 33 can also exclude the court martial (tribunals established under the military law) from the writ jurisdiction of the Supreme Court and the high courts, so far as the enforcement of Fundamental Rights is concerned. **Hence statement 3 is correct.**

- Q 11. D**
- The Balance of Payments (BoP) is a systematic record of all economic transactions between a country's residents (individuals, businesses, and government) and the rest of the world over a specific period, typically a quarter or a year. It summarizes inflows (credits) and outflows (debits) of funds, crucial for assessing a nation's economic health, international trade competitiveness, and currency stability.
 - Accommodating transactions (**termed 'below the line items'**) refer to transactions that take place to cover deficit (or surplus) arising from autonomous transactions. They are determined by the gap in the balance of payments, that is, whether there is a deficit or surplus in the balance of payments. Since the official reserve transactions are made to bridge the gap in the BoP, they are seen as the accommodating item in the BoP (all others being autonomous). In simple terms, If there's a gap, the central bank uses reserves, loans, or foreign exchange operations to "accommodate" it. **Hence option (d) is the correct answer.**

- Q 12. C**
- The Direct Firing Sight Navigation System is a combat technology package that allows artillery units to engage targets directly and autonomously. The system integrates the CM3-MR Direct Firing

Sight, which utilises advanced optical and thermal technologies to significantly enhance accuracy during line-of-sight (direct-fire) engagements. Hence, statement 1 is correct.

- **It incorporates the SIGMA 30N inertial navigation system.** Unlike standard GPS, this inertial system calculates positions using motion sensors, allowing it to operate autonomously in GPS-denied or heavily jammed electronic warfare environments. **Hence, statement 2 is correct.**
- The system integrates two critical components: SIGMA 30N Navigation System and CM3-MR Direct Firing Sight.
- The SIGMA 30N is an advanced inertial navigation system used for autonomous artillery positioning and targeting. The high-precision sighting and real-time thermal optics make the system particularly lethal against modern battlefield threats, including low-flying drones, fast-moving vehicles, and other visible targets. **Hence, statement 3 is correct.**

- Q 13. C**
- Recently, the Supreme Court has clarified that any inclusion or exclusion of any caste, race or tribe or part of or group within the castes, races or tribes has to be, by law made by the Parliament, and not by any other mode or manner.
 - **Clause (1) of Article 341 mandates that the President, after consultation with the State Governor, may by public notification specify the castes, races or tribes or parts of or groups within castes, races or tribes which shall for the purposes of this Constitution be deemed to be Scheduled Castes in relation to a State or Union Territory.**
 - **Clause (2) adds that any amendment or alteration or variation to the President's List under Article 341 have to be made by a parliamentary law alone, and not by any notification.**
 - **The Supreme Court concluded that the State government had no competence/authority/power to tinker with the lists of Scheduled Castes published under Article 341 of the Constitution. Hence, option (c) is the correct answer.**

- Q 14. A** India added Gogabeel Lake in Bihar, Siliserh Lake in Rajasthan and Kopra reservoir in Chhattisgarh to the List of Wetlands of International Importance (the "Ramsar List"). India now has 96 wetlands under the Ramsar Convention.

- **Siliserh Lake:** It is in the semi-arid zone and falls in the buffer region of the Sariska tiger reserve. It is a human-made wetland, created in 1845 AD by Maharaja Vinay Singh by creating a bund across the tributary of the River Ruparel. Biodiversity: Egyptian Vulture, Black Stork, Tiger, Indian Pangolin, Leopard and Sambar. **Hence, point 3 is correct.**
- **Kopra Jalashay:** Located in the Bilaspur district, this reservoir is significant for having nutrient-rich backwaters and serving as the first recognised, high-value wetland in Chhattisgarh. It is a reservoir located in the upper catchments of the River Mahanadi. Its extensive open water area features shallow, nutrient-rich backwaters.
- **Gogabeel Lake:** It is an oxbow wetland situated in Katihar district, Bihar. Gogabeel is formed from the flow of the rivers Mahananda and Kankhar in the north and the Ganga in the south and east. It is the first community reserve of Bihar. It is a habitat for a wide variety of birds some of which are even migratory birds. **Hence, point 1 is correct.**
- **Longwood Shola Reserve Forest (Tamil Nadu):** Added in 2023–2024, this site is part of the high-altitude evergreen forests in the Nilgiris, crucial for water security and unique biodiversity. **Hence, point 2 is not correct.**

- Karaivetti Bird Sanctuary (Tamil Nadu): Added in 2023–2024, it is one of the largest inland freshwater lakes in Tamil Nadu and a critical stopover on the Central Asian Flyway. **Hence, point 4 is not correct.**

- Q 15. A**
- Biotic potential is the maximum, theoretical rate at which a population increases under ideal environmental conditions—unlimited resources, no predators, and no diseases. It represents the highest possible birth rate and lowest mortality rate for a species. Key factors include frequency of reproduction, number of offspring, and maturity age.
 - Full expression of the biotic potential of an organism is restricted by environmental resistance, any factor that inhibits the increase in the number of the population. These factors include unfavourable climatic conditions; lack of space, light, or a suitable substrate; deficiencies of necessary chemical compounds or minerals; and the inhibiting effects of predators, parasites, disease organisms, or unfavourable genetic changes.
 - **Hence, option (a) is the correct answer.**

- Q 16. C**
- One of the key components of the National Rural Health Mission is to provide every village in the country with a trained female community health activist ASHA or Accredited Social Health Activist. Selected from the village itself and accountable to it, the ASHA will be trained to work as an interface between the community and the public health system.
 - She should be a literate woman with due preference in selection to those who are qualified up to 10 standard wherever they are interested and available in good numbers. This may be relaxed only if no suitable person with this qualification is available.
 - The ASHAs will receive performance-based incentives for promoting universal immunization, referral and escort services for Reproductive & Child Health (RCH) and other healthcare programmes, and construction of household toilets.
 - Hence, only statement 2 & 4 are correct.

Q 17. D Panel of chairpersons of Lok Sabha

- **Under the Rules of Lok Sabha(not Constitution), the Speaker nominates from amongst the members a panel of not more than ten chairpersons. Hence statement 1 is not correct.**
- Any of them can preside over the House in the absence of the Speaker or the Deputy Speaker. He/she has the same power as the Speaker when so presiding. **It must be emphasised here that a member of the panel of chairpersons cannot preside over the House, when the office of the Speaker or the Deputy Speaker is vacant.** During such time, the Speaker's duties are to be performed by such member of the House as the President may appoint for the purpose. The elections are held, as soon as possible, to fill the vacant posts. **Hence statement 2 is not correct.**
- The members has the same power as the Speaker when so presiding. He/she holds office until a new panel of chairpersons is nominated. **When a member of the panel of chairpersons is also not present, any other person as determined by House acts as the Speaker. Hence statement 3 is not correct.**

- Q 18. D** • The Aravalli Range is one of the oldest geological features in the world, dating back to the Precambrian era. They were originally "Fold Mountains" (formed by the folding of the Earth's crust), much like the Himalayas are today. However, due to millions of years of weathering and erosion, they are now classified as Relict or Residual Mountains. **Hence, statement 1 is correct.**

The Northern Western Ghats (specifically the Sahyadris in Maharashtra) are indeed part of the Deccan Traps.

- During the Mesozoic era, massive volcanic eruptions covered this region in thick layers of basaltic lava.
- Geologically, the Western Ghats are not "true" mountains formed by folding; they are the faulted edge (escarpment) of the Peninsular Plateau that remained after the western portion subsided into the Arabian Sea. **Hence, statement 2 is correct.**
- The Meghalaya Plateau (comprising the Garo, Khasi, and Jaintia Hills) is geologically a part of the Peninsular Shield.
- During the Himalayan orogeny, a massive force caused a portion of the Peninsular block to sink.
- This created a downward gap known as the Malda Gap (or Garo-Rajmahal Gap), which separated the Meghalaya Plateau from the main Chota Nagpur Plateau.
- Over time, this gap was filled by the sediments of the Ganga and Brahmaputra rivers, forming the Bengal Delta, but the underlying basement rock of Meghalaya remains Peninsular in origin. **Hence, statement 3 is correct.**

- Q 19. B** • Temperature inversion is a meteorological phenomenon where temperature increases with altitude, reversing the normal lapse rate. For this to occur near the Earth's surface, specific conditions must facilitate rapid cooling of the ground and prevent the mixing of air layers:

Clear and cloudless long winter nights:

- Long Nights: During winter, nights are longer than days. This ensures that the period of outgoing terrestrial radiation is significantly longer than the period of incoming solar radiation, allowing the surface to lose more heat.
- Clear Skies: Clouds act like a greenhouse "blanket" that traps and re-radiates terrestrial radiation back to the surface. Clear, cloudless skies allow heat to escape unhindered into space, leading to rapid surface cooling. **Hence, statement 1 is correct.**

Dry air near the Earth's surface:

- Water vapor is a potent greenhouse gas that absorbs infrared radiation (heat).
- Dry air has a low heat-capacity and allows outgoing terrestrial radiation to pass through more easily. This facilitates the rapid cooling of the ground, which then chills the layer of air immediately above it. **Hence, statement 2 is correct.**

Strong surface winds:

- Temperature inversion requires calm or stable air.
- Strong winds cause mechanical turbulence and vertical mixing. This mixing forces the chilled air at the surface to swap places with the warmer air above, effectively preventing a distinct "cold layer" from forming at the bottom. **Hence, statement 3 is not correct.**

Strong convective activity during night-time:

- Convection is the process where warm air rises and cool air sinks.

- Strong convective activity implies that air is moving vertically. For an inversion to occur, the atmosphere must be stable, meaning the heavy, cold air needs to remain "trapped" at the bottom. Convective activity would break this stability and mix the layers. **Hence, statement 4 is not correct.**

- Q 20. A**
- Recent Context: The Carney doctrine was proposed by Canadian Prime Minister Mark Carney during a speech at the World Economic Forum in Davos, Switzerland, on January 20, 2026.
 - The Carney Doctrine, introduced by Canadian Prime Minister Mark Carney in early 2026, is a foreign policy framework urging middle powers to unite against coercive actions by great powers. It advocates for economic diversification, reduced reliance on singular superpowers, and strengthening international rules-based systems through collective action rather than passive accommodation. **Hence option (a) is the correct answer.**
 - Mark Joseph Carney is a Canadian politician and economist who has served as the 24th prime minister of Canada since 2025.

- Q 21. C**
- Laser light communication (free-space optics) is a wireless, high-bandwidth technology that transmits data using infrared laser beams, acting as a faster, more secure alternative to RF systems. It enables, for example, 10 to 100 times higher data rates for space-to-ground communication.
 - Laser light is inherently coherent, meaning the light waves are in phase both spatially and temporally. This allows for the creation of highly focused, narrow beams that can transmit data over long distances with minimal divergence. **Hence, statement 1 is correct.**
 - Because lasers operate at much higher frequencies (terahertz range) than radio waves (kilohertz to gigahertz range), they can carry significantly higher bandwidth, enabling the transfer of much more data per second. **Hence, statement 2 is not correct.**
 - This technology is actively used and developed for satellite-to-satellite (inter-satellite) links and deep space missions. For example, NASA's DSOC (Deep Space Optical Communication) has successfully demonstrated laser communication from millions of kilometres away. **Hence, statement 3 is correct.**

- Q 22. B**
- **Refuse Derived Fuel (RDF)** is a high-calorific fuel produced from processed municipal solid waste, industrial, or commercial waste, including plastics, paper, and textiles. It serves as an eco-friendly alternative to fossil fuels, commonly used in cement kilns and power plants, reducing landfill reliance.
 - RDF can be further specified into- Tyre Derived Fuels (TDF), Solid Recovered Fuels (SRF), and Alternative Fuels (RDF). RDF consists mainly of combustible waste, including non-recyclable plastics (except PVC), paper, cardboard, labels, and other corrugated materials.
 - Dumping of waste in low-lying areas is unscientific and filthy, which leads to methane emissions, water pollution, and soil degradation. In contrast, this renewable energy source is an eco-friendly option and a viable landfill alternative. The conversion of MSW to RDF contributes to the scientific management of MSW. It also replaces the use of fossil fuels. The fuel is derived after basic processing to increase the calorific value of Combustible waste materials.
 - **Hence, option (b) is the correct answer.**

Q 23. B Tamil Nadu IAS officer Supriya Sahu was awarded for her pioneering leadership on critical environmental challenges in India, including plastics and wildlife conservation.

- The UNEP has administered the award annually since its inception in 2005 to recognize significant contributions to environmental protection, climate action, and sustainable development. **Hence, statement 1 is correct.**
- The Champions of the Earth award is the UN's highest environmental honour. Every year since 2005, UNEP honours individuals and organizations working on innovative and sustainable solutions to address the triple planetary crisis of climate change, nature and biodiversity loss, and pollution and waste. **Hence, statement 2 is correct.**
- Supriya Sahu was honored in the "Inspiration and Action" category in 2025 for her work in Tamil Nadu, including ecosystem restoration and sustainable cooling. Supriya Sahu is not the only Indian to receive this award. Several other Indians have received it previously, including Prime Minister Narendra Modi (2018), Afroz Shah (2016), and Purnima Devi Barman (2022). **Hence, statement 3 is not correct.**

Q 24. A • During the sixteenth century, known as the Vijayanagara Period in Andhra Pradesh, the sculptors experimented with portrait sculpture in order to preserve knowledge of the royal patron for posterity. **At Tirupati, life-size standing portrait statues were cast in bronze, depicting Krishnadevaraya with his two queens, Tirumalamba and Chinnadevi.** The sculptor has combined the likeness of the facial features with certain elements of idealisation. **Hence, option (a) is correct.**

Q 25. B • Report published by the UN Environment Programme's (UNEP) Cool Coalition, details a Sustainable Cooling Pathway aimed at dramatically reducing projected GHG emissions from cooling by 2050.

- The Global Cooling Watch 2025 report, titled "The Free Degrees," was released by the United Nations Environment Programme (UNEP) and the Cool Coalition on November 11, 2025, at the COP30 summit in Belém, Brazil.

Key concerns raised in the Report

- Surge in cooling demand: Under a Business-as-Usual (BAU) Cooling Pathway, the global stock of cooling equipment will more than triple by 2050, from 22 terawatts (TW) in 2022 to 68 TW.
- Policy Gaps: Only 54 nations meet full Sustainable Cooling Pathway standards despite broad policy inclusion.
- Escalating extreme heat: Global population exposed to deadly heat stress could rise from 30% today to between 48% and 76% by the end of the century. (IPCC)
- This gets further amplified by the urban heat island effect, heatwaves, etc.

Proposed Sustainable Cooling Pathway

- Passive Cooling: Reduce cooling loads through passive design, urban planning, and simple measures like doors on refrigerated cabinets to cut costs and emissions.
- Low-Energy Cooling: Prioritise low-energy systems such as fans and evaporative coolers instead of, or alongside, air conditioning to cut energy use and costs.
- Best Energy Efficiency: Adopt high-efficiency systems with variable-speed compressors and proper maintenance for optimal performance.

- Rapid HFC Phase-Down: Select low-GWP refrigerants that maintain system efficiency to reduce direct emissions and support sustainable cooling.
- **Hence, option (b) is the correct answer.**

Q 26. B • Mariana Trench is located in the Western Pacific Ocean, it contains the Challenger Deep, which is the deepest known point in the world's oceans (approx. 10,935 meters). **Hence, pair 1 is correctly matched.**

- Amazon River is a classic geographic debate, but according to the Guinness World Records and most geographers, the Nile River is the longest in the world by length. The Amazon is the largest river in the world by volume of water discharge and possesses the largest drainage basin, but it ranks second in total length. **Hence, pair 2 is not correctly matched.**
- Lake Superior is located on the border of the USA and Canada, it is the largest freshwater lake by surface area (82,103 km²).
- Note: If the question asked for the largest freshwater lake by volume, the answer would be Lake Baikal in Russia. **Hence, pair 3 is correctly matched.**

Q 27. B **The National Stock Exchange (NSE), established in 1992, revolutionized Indian capital markets by being the first to introduce a fully automated, screen-based electronic trading system in 1994.**

- This significantly improved transparency, speed, and efficiency in trading operations, replacing the old open outcry system that was used at BSE and other regional exchanges.
- **Hence statement 1 is correct.**

There is no legal or regulatory requirement mandating that companies, large-cap or otherwise, must list on both NSE and BSE. A company can choose to be listed on only one of the two exchanges.

- However, many large-cap companies voluntarily list on both to increase liquidity and accessibility for investors, but it is a business decision, not a mandate.
- **Hence statement 2 is not correct.**

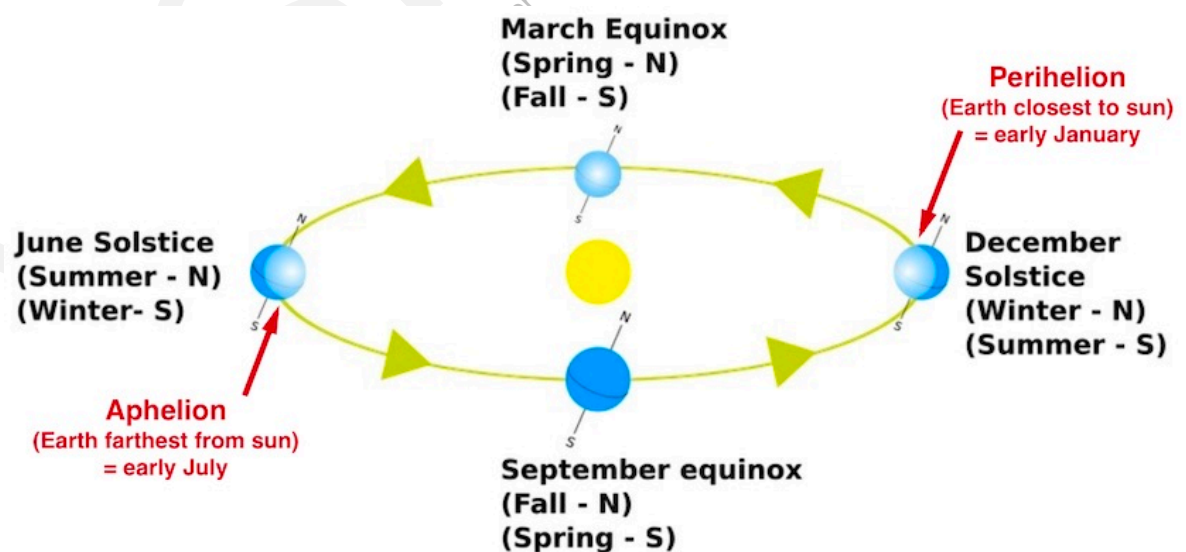
The India International Exchange (India INX) is a wholly owned subsidiary of the Bombay Stock Exchange (BSE).

- It was launched in 2017 at GIFT City, Gujarat, and serves as India's first international stock exchange, offering products such as equities, derivatives, commodities, and debt instruments.
- It was set up to position India as a hub for international financial services.
- **Hence statement 3 is correct.**
- BSE, the first ever stock exchange in Asia was established in 1875 and the first in the country to be granted permanent recognition under the Securities Contract Regulation Act, 1956.

Q 28. C • Recently, the **Chief Election Commissioner highlighted the need to facilitate postal voting for non-resident Indians (NRIs)**. In this regard, the Election Commission's proposal for an **Electronically Transmitted Postal Ballot System (ETPBS)** for NRIs, which is currently awaiting government approval.

- ETPBS is the system through which **individual voters cast their vote on an electronically received postal ballot**, from anywhere outside their constituency. For this, it combines the use of **electronic transmission** and **postal services** to enable voters to receive and return their ballots electronically. **Hence, statement 1 is correct.**
- This is a **fully secured system with two security layers.**
- The **Conduct of Election Rules, 1961**, was **amended in 2016** to allow only "**service voters**" to use the ETPBS. **Hence, statement 2 is correct.**
- A **service voter** is a voter with service qualifications. According to the provisions of sub-section (8) of Section 20 of **Representation of People Act, 1950**, service qualification means:
 - (a) Being a member of the **armed Forces** of the Union ; or
 - (b) Being a member of a force to which provisions of the **Army Act, 1950** (46 of 1950), have been made applicable, whether with or without modification;
 - (c) Being a member of an **Armed Police Force** of a state and serving outside that state; or
 - (d) Being a person who is **employed under the Government of India, in a post outside India.**
- The same system is proposed to be extended for the NRIs.

- Q 29. B**
- The Earth's axis is tilted at an angle of 23.5° from its orbital plane. As the Earth revolves around the Sun, this tilt causes different parts of the Earth to receive varying amounts of direct sunlight at different times of the year. This variation in solar intensity and day length is the primary cause of the season. **Hence, statement 1 is correct.**
 - Perihelion is the point where Earth is closest to the Sun, not farthest. It occurs around January 3rd each year. The point where Earth is farthest from the Sun is called Aphelion. **Hence, statement 2 is not correct.**
 - Aphelion is the point where Earth is farthest from the Sun (approximately 152 million km). This occurs annually around July 4th. During this time, the Northern Hemisphere is tilted toward the Sun, which is why it coincides with the Northern Hemisphere summer. **Hence, statement 3 is correct.**



Q 30. A Servants of India Society

In 1905, the Servants of India Society was established by **Gopal Krishna Gokhale in Pune**. He was a prominent moderate leader of the Indian National Congress. Gopal Krishna Gokhale gathered a few

volunteers who shared his vision of selfless service. The foundation marked a new approach to nationalism rooted in education, social upliftment, and sacrifice.

Servants of India Society Objectives

The aims of the Servants of India Society were far-reaching and struck deep roots in social reform. Their main objectives were:

- Empower the poor and marginalised through education.
- Encouragement of public health, eradication of disease.
- Fighting social evils such as caste discrimination and untouchability.
- Strengthening national unity by diffusing national consciousness.
- Social service in support of rural development and self-reliance.
- Training people to give themselves to service rather than to personal interests.
- These objectives aimed at bringing forth a generation of "servants" that put the welfare of the public first and above their personal ambitions.

Hence, option (a) is the correct answer.

Q 31. B Indian researchers have identified biochemical markers in the blood that could help detect kidney complications in diabetic patients.

Biochemical markers are measurable indicators in biological samples (blood, urine, tissues) that reflect normal or abnormal biological states, such as disease presence, progression, or treatment response, including substances like proteins, enzymes (e.g., troponin for heart attack), lipids, or metabolites (e.g., lactate). Used in medicine, genetics, and toxicology, they help diagnose conditions (like cancer tumor markers), monitor organ function (e.g., liver enzymes), assess risk (high cholesterol), or study genetic variation (isozymes).

Key Types & Examples

- Diagnostic Markers: Indicate specific diseases.
- Cardiac: Troponin, CK-MB for heart attack.
- Cancer: PSA (prostate), CEA (colon).
- Metabolic: Glucose, HbA1c for diabetes.
- Prognostic/Predictive Markers: Show disease severity or predict treatment response (e.g., certain proteins in cancer).
- Monitoring Markers: Track organ health or response to therapy.
- Liver: ALT, AST.
- Bone Turnover: ALP, CTX.
- Genetic Markers: Variations in proteins (isozymes) used in breeding/genetics.

Hence, option (b) is the correct answer.

Q 32. D • In India, political parties are classified into Recognised Political Parties and Registered Unrecognised Political Parties (RUPP) under the provision of the Representation of the People Act, 1951.

- The Registered Unrecognised Political Parties (RUPP) are registered with the ECI but they do not meet the criteria to be recognized as either a State or National party.

- The RUPPs cannot use a reserved election symbol but may apply for a common symbol under certain conditions. As per the ECI, common symbols are provided to RUPPs based upon an undertaking that they would put up at least 5% of total candidates with regard to said Legislative Assembly election of a State. Hence, statement (1) is not correct.
- As per Section 29C of RPA, 1951 requires RUPPs to furnish a contribution report under Conduct of Election Rules, 1961. Such contributions received by RUPPs are also 100% exempted from Income Tax as an incentive to the parties to strengthen the electoral democracy. Hence, statement (2) is not correct.
- The RPA, 1951 provides that a recognized political party (National or State) can appoint a maximum of 40-star campaigners while Registered Unrecognised Political Parties can appoint up to 20. Hence, statement 3 is not correct.

- Q 33. B**
- Based on scientific consensus and atmospheric studies, the impacts of stratospheric ozone depletion are as follows:
 - Increased UV-B radiation due to ozone depletion harms plant life by damaging DNA and photosynthetic machinery. This leads to stunted growth, reduced biomass, and changes in phenological stages (timing of flowering and development). **Hence, point 1 is correct.**
 - Thinner stratospheric ozone allows more UV radiation to reach the surface, which is directly linked to higher rates of skin cancer, cataracts, and immune system suppression in humans. **Hence, point 3 is correct.**
 - Increased UV-B radiation penetrating the lower atmosphere enhances the photodissociation of trace gases, which accelerates the chemical reactions that produce ground-level ozone (a component of smog). **Hence, point 4 is correct.**
 - Ozone is a greenhouse gas that absorbs solar radiation, warming the atmosphere. Its depletion leads to cooling of the stratosphere, not the troposphere. In fact, the overall effect of ozone depletion and the greenhouse gases that cause it (like CFCs) is generally associated with global warming of the troposphere (lower atmosphere). **Hence, point 2 is not correct.**

Q 34. A **Recent Context: The 2026 Commonwealth Games (XXIII Commonwealth Games), or Glasgow 2026, will be held in Glasgow, Scotland, from 23 July to 2 August 2026**

Gabon and Togo were the last two countries to join the Commonwealth in 2022, highlighting its evolving membership and continued relevance as a voluntary association of independent states.

Topic in Focus:

The Commonwealth is a political association of 56 member states, most of which are former territories of the British Empire. It is a voluntary organization united by language, history, culture, and shared values of democracy, human rights, and the rule of law. The Commonwealth aims to foster international cooperation and trade, promoting development and understanding among its diverse members across the globe.

Key Aspects:

- **Genesis and Evolution:** Evolved from the British Empire, emphasizing equality among members.
- **Head of Commonwealth:** British monarch is symbolic head, not ruler of member states.
- **London Declaration (1949):** Allowed republics to retain membership within the association.
- **Membership Criteria:** Voluntary, open to countries with historical links to the UK.
- **Commonwealth Secretariat:** Central intergovernmental organization managing its work.

Analysis of Statements:

Statement 1: Incorrect. Member countries owe allegiance to the British monarch as the symbolic Head of the Commonwealth, not as their ruler.

Statement 2: Correct. The London Declaration (1949) enabled republics to remain members, changing the nature of allegiance.

Statement 3: Incorrect. The Commonwealth Secretariat was established in 1965, not in 1947. The Commonwealth Secretariat is the principal intergovernmental agency of the Commonwealth of Nations, established in 1965 and headquartered in London.

Hence, the correct answer is option (a).

- Q 35. D** The Gupta kingdom was enlarged enormously by Chandragupta's son and successor Samudragupta (AD 335–80). **He was the opposite of Ashoka. Ashoka believed in a policy of peace and non-aggression, but Samudragupta delighted in violence and conquest.** His court poet Harishena wrote a glowing account of the military exploits of his patron, and, in a long inscription, the poet enumerate the peoples and countries that were conquered by Samudragupta. The inscription is engraved at Allahabad on the same pillar that carries the inscriptions of the peace-loving Ashoka. **Hence statement 1 is not correct.**

The reign of Chandragupta II (AD 380-412) saw the high watermark of the Gupta empire. He extended the limits of the empire by marriage alliance and conquest. Chandragupta II adopted the title of Vikramaditya, which had been first used by an Ujjain ruler in 58–57 BC as a mark of victory over the Shaka Kshatrapas of western India. **It was during Chandragupta's reign that the Chinese pilgrim Fa-hsien (AD 399–414) visited India and wrote an elaborate account of the life of its people. Hence statement 2 is not correct.**

- Q 36. C**
- **Black carbon:** Black carbon is a major component of fine particulate matter (PM_{2.5}) produced by the incomplete combustion of fossil fuels (particularly diesel in transport) and biofuels/biomass (used for cooking and heating). **Hence, Pair 1 is correctly matched.**
 - **Arsenic:** Phosphate rocks used to manufacture fertilisers naturally contain various levels of arsenic as an impurity. Long-term application of these fertilisers is a known anthropogenic source of arsenic accumulation in soil and groundwater. **Hence, Pair 2 is correctly matched.**
 - **Mercury:** Coal-fired power plants are the largest anthropogenic source of mercury emissions globally. Mercury is naturally present in coal and is released into the atmosphere during combustion. **Hence, Pair 3 is correctly matched.**
 - **Plutonium:** Plutonium is a heavy synthetic element primarily produced in nuclear reactors as a byproduct of uranium irradiation. It is used in nuclear weapons and as fuel in some nuclear power reactors. Medical radioisotopes typically involve lighter isotopes (like Technetium-99m or Iodine-131) and do not involve the production of plutonium. **Hence, Pair 4 is not correctly matched.**

- Q 37. C**
- A **liquidity trap** is a contradictory economic situation in which **interest rates are very low** and **savings rates are high**, rendering monetary policy ineffective. **Hence option (c) is the correct answer.**
 - It was first described by economist John Maynard Keynes.
 - During a liquidity trap, consumers choose to avoid bonds and keep their funds in cash savings because of the prevailing belief that interest rates could soon rise (which would **push bond prices down and yields up**). At the same time, central bank efforts to spur economic activity are

hampered as they are unable to lower interest rates further to incentivize investors and consumers.

- While a liquidity trap is a function of economic conditions, it is also psychological since consumers are making a choice to hoard cash instead of choosing higher-paying investments because of a negative economic view.
- A liquidity trap is not limited to bonds. It also affects other areas of the economy, as consumers are spending less on products which means businesses are less likely to hire. Some ways to get out of a liquidity trap include raising interest rates, hoping the situation will regulate itself as prices fall to attractive levels, or increased government spending.
- **Inflationary Paradox:** This contradictory situation, where consumer spending increases despite high inflation, is often referred to as an "inflationary paradox" or "paradox of high inflation and high spending". "Buy Now" Mentality (Hedonic Treadmill): Consumers expect prices to rise even higher in the future, prompting them to purchase goods immediately before they become more expensive.

Q 38. D Consumer Price Index (CPI)

- The Consumer Price Index (CPI) assigns weights to different groups of goods and services based on their share in household consumption expenditure.
- These weights indicate the relative importance of each category in determining overall retail inflation.

Major Groups in CPI Basket

The CPI basket is divided into the following broad groups:

- Food and Beverages
- Pan, Tobacco and other Intoxicants
- Clothing and Footwear
- Housing
- Fuel and Light
- Miscellaneous (services and other consumer items)

Key Weightage Insights (CPI–Combined)

Food and Beverages

- Highest weightage in CPI
- Accounts for 45.86%
- Reflects the dominance of food expenditure in household consumption, especially in rural India

Pan, Tobacco and Other Intoxicants

- Lowest weightage in CPI at 2.38%
- Earlier included under Food and Beverages
- Now treated as a separate group.
- **Hence, option 3 is correct.**

Miscellaneous

- Second-highest weightage at 28.32%
- Includes services such as health, education, transport, communication, and personal care

Housing

- Weightage of 10.07%
- Applicable only to urban CPI
- Excluded from rural CPI due to different housing ownership patterns.
- **Hence, option 4 is correct.**

Fuel and Light

- Accounts for 6.84%
- Includes electricity, firewood, LPG, kerosene, etc.
- **Hence, option 2 is correct.**

Clothing and Footwear

- Weightage of 6.53%
- Lower than Fuel and Light, making it one of the lower-weight groups in CPI.
- **Hence, option 1 is correct.**

Q 39. B National Commission for Backward Classes (NCBC) was set up in 1993. Later, the 102nd Amendment Act of 2018 conferred a constitutional status on the Commission. Hence statement 1 is not correct.

- For this purpose, the amendment inserted a new Article 338-B in the constitution. Hence, the Commission ceased to be a statutory body and became a constitutional body.

The Commission consists of a chairperson, a vice-chairperson and three other members. They are appointed by the President by warrant under his hand and seal. Their conditions of service and tenure of office are also determined by the President.

- **Hence statement 3 is correct.**

The functions of the Commission are the following :

- To investigate and monitor all matters relating to the constitutional and other legal safeguards for the socially and educationally backward classes and to evaluate their working.
- To inquire into specific complaints with respect to the deprivation of rights and safeguards of the socially and educationally backward classes.
- To participate and advise on the socioeconomic development of the socially and educationally backward classes and to evaluate the progress of their development under the Union or a state.
- To present to the President, annually and at such other times as it may deem fit, reports upon the working of those safeguards.
- To make recommendations as to the measures that should be taken by the Union or a state for the effective implementation of those safeguards and other measures for the protection, welfare and socio-economic development of the socially and educationally backward classes.
- To discharge such other functions in relation to the protection, welfare, development and advancement of the socially and educationally backward classes as the President may specify.

The Commission presents an annual report to the President. It can also submit a report as and when it thinks necessary.

- The President places all such reports before the Parliament, along with a memorandum explaining the action taken on the recommendations made by the Commission.
- **Hence statement 2 is correct.**

- The Commission presents an annual report to the President. It can also submit a report as and when it thinks necessary. The President places all such reports before the Parliament, along with a memorandum explaining the action taken on the recommendations made by the Commission

Q 40. A • Lake Turkana is recognised as the world's largest permanent desert lake and the world's largest alkaline lake. Its harsh, arid environment earns it the nickname "The Jade Sea" due to its stunning turquoise colour. **Hence, statement 1 is correct.**

- It is a tectonic lake situated in the Eastern Branch of the East African Rift System. The basin was formed by the pulling apart of the African tectonic plate, which created the depression that holds the lake today. **Hence, statement 2 is correct.**
- Lake Turkana is not located entirely within Ethiopia. While its primary northern tributary (the Omo River) flows from Ethiopia, the vast majority of the lake lies within Kenya. Only the northernmost tip of the lake crosses the border into Ethiopia. **Hence, statement 3 is not correct.**

Q 41. A • In a weight-losing industry, the weight of the raw material is higher than the weight of the finished product (e.g., Sugar industry, where 100 kg of sugarcane yields only about 10–12 kg of sugar). To minimize the cost of transporting bulky, "gross" raw materials that will eventually be discarded as waste, the factory is placed near the source. **Hence, statement 1 is correct.**

- Footloose industries (like software development, mobile assembly, or diamond cutting) are characterized by producing high-value, low-bulk products. They are not tied to any specific raw material source because their inputs and outputs are easy to transport. Instead, they are influenced by factors like accessibility, availability of skilled labor, and telecommunications. **Hence, statement 2 is not correct.**
- If a raw material is highly perishable (like milk for a milk-processing plant or grapes for a winery), the industry must be located close to the raw material source. Delay in transporting perishable items to a distant market-based factory would lead to the spoilage of the inputs. (Note: Industries where the final product is perishable, like a bakery, tend to be market-oriented). **Hence, statement 3 is not correct.**

Q 42. A • Quantum dots are indeed semiconductor nanoparticles (nanocrystals) with dimensions typically ranging from 2 to 10 nanometers. At this scale, they exhibit properties that are intermediate between bulk semiconductors and discrete atoms. **Hence, statement 1 is correct.**

- The defining feature of quantum dots is the quantum confinement effect. The emission wavelength (colour) is strictly dependent on the particle size. Smaller quantum dots have a larger bandgap and emit shorter wavelengths (blue light), while larger dots have a smaller bandgap and emit longer wavelengths (red light).
- No solar cell can reach a 100% thermodynamic efficiency. While quantum dot solar cells are researched for their ability to exceed the Shockley-Queisser limit (approx. 33.7% for single junctions) through processes like Multiple Exciton Generation (MEG), the absolute thermodynamic limit for any solar energy conversion (the Landsberg limit) is approximately 93-95%. In practice, quantum dot solar cell efficiencies are currently significantly lower than this theoretical maximum. **Hence, statement 2 is not correct.**

Q 43. A • **The Tillana, corresponding to the Tarana of Hindustani music, is a short and crisp form. It is mainly a dance form,** but on account of its brisk and attractive music, it sometimes finds a place in music concerts as a conclusion piece. It usually begins with jatis. **Hence, statement 1 is correct.** The name Tillana is constituted of the rhythmic syllables, ti la na. It is the liveliest of musical forms. This form is said to have had its birth in the 18th century. **The sahitya of a tillana may be in Sanskrit, Telugu or Tamil.** The presence of rhythmical solfa syllables alongwith a sprinkling of sahitya enhance the beauty of the form of the Tillana. The music is of comparatively slow tempo in Tillanas meant for dance purposes. The pallavi and anupallavi consists of jatis and the charana has sahitya, jatis and svaras. Ramnad Srinivasa Iyengar, Pallavi Seshayyar and Swati Tirunal are some of the prominent composers of Tillanas. **Hence, statement 2 is not correct.**

Q 44. C Planetary winds are large-scale air masses that blow relatively consistently across the globe throughout the year. Their primary formation is driven by three main factors:

- **Unequal heating (Latitudinal variation):** The Sun heats the Earth unevenly, with the equator receiving more direct radiation than the poles. This temperature difference creates air movement as warm air rises and cool air sinks.
- **Pressure Belts:** This differential heating leads to the development of permanent high and low-pressure belts (such as the Equatorial Low or Subtropical High). Wind naturally flows from high-pressure areas to low-pressure areas.
- **Coriolis Force:** Caused by the Earth's rotation on its axis, this force deflects the path of the wind (to the right in the Northern Hemisphere and left in the Southern Hemisphere), giving planetary winds like the Trade Winds and Westerlies their characteristic directions.
- **While the Earth's revolution around the Sun causes seasonal shifts** (migration) of these wind belts, it does not play a direct role in the fundamental formation of the planetary wind system itself.
- **Hence, option (c) is the correct answer.**

Q 45. B **Context: The RBI has recently set the WMA limit for the Central Government for the second half of the 2025–26 financial year. WMA limit is decided by RBI in consultation with the Central government.**

About WMA

- It is a temporary advance given by the RBI to the central to cover any mismatch in receipts and payments.
- Unlike ad hoc Treasury Bills, the instrument of WMAs is not a source of financing fiscal deficit.
- WMA is also available for state governments and Union Territories.
- Legal framework: RBI Act, 1934 authorises the central bank for WMA
- Interest rate: Interest rate on WMA is the RBI's repo rate
- However, interest on overdraft (amounts in excess of their WMA limits) is 2 % above the repo rate.
- Duration- It shall be fully paid off within 3 months from the date of making the WMAs.

Hence, option (b) is the correct answer.

Q 46. C • Article 224 of the Indian Constitution deals with appointing additional and acting judges in high courts. The provision allows for the appointment of judges to fill vacancies temporarily or to deal

with an increase in the workload of a particular court. According to Article 224, if at any time the number of judges of a high court is insufficient to carry out its work, the Chief Justice of the high court may, with the previous consent of the President, request the Governor of the state to appoint additional judges for a temporary period. These additional judges have the same powers, functions, and privileges as the permanent judges of the high court. There is no such provision for the Supreme Court. **Hence statement 1 is correct.**

- President, in consultation with the chief justice of India, has the final power to determine the age of the high court judge if any question is raised by anybody on that behalf under article 217, while under article 124, the age of a supreme court shall be determined in such a manner as the president by law provides. **Hence statement 2 is correct.**
- The President may transfer a Judge from one High Court to another High Court after consultation with the Chief Justice of India. On transfer, the judge of the High Court is entitled to receive an addition to his salary, such as a compensatory allowance as may be determined by Parliament. **Hence statement 3 is correct.**

Q 47. C • Induced pluripotent stem cells (iPSCs) are a cornerstone of regenerative medicine. They are created by reprogramming adult cells (like skin or blood cells) back into an embryonic-like state, allowing them to differentiate into any cell type to repair or replace damaged tissues. **Hence, statement I is correct.**

- iPSCs are not naturally present in the body or placental tissues. They are man-made (induced) in a laboratory using genetic reprogramming factors (such as the Yamanaka factors). While placental tissues and umbilical cord blood contain multipotent stem cells (like mesenchymal or hematopoietic stem cells), they do not naturally contain pluripotent iPSCs. **Hence, statement II is not correct.**

Q 48. A • Exfoliation domes are large, smooth, rounded domes of bedrock, and the process that creates them, exfoliation or sheeting, is most common in

- massive, coarse-grained, homogeneous, intrusive igneous rocks like granite. Because these rocks lack internal joints or bedding planes, the stress relief due to unloading occurs along fractures parallel to the ground surface, leading to the characteristic sheet-like peeling. **Hence, statement I is correct.**
- **Statement-II** accurately describes the primary mechanism of exfoliation, known as unloading or pressure release. Granite forms deep beneath the Earth's surface under tremendous pressure from the overlying rock (the overburden). As erosion strips away this overburden, the rock body expands slightly outward. This expansion causes concentric fractures (called joints) to form parallel to the exposed surface, leading to the separation and peeling of the outer layers.
- **Hence**, Statement-II explains the physical mechanism (reduction in pressure causing expansion and separation) that leads to the formation of the large, dome-like landforms (exfoliation domes) commonly seen in rocks like granite (Statement-I).

Q 49. B • **The Green Gross Domestic Product**, or Green GDP is an indicator of economic growth with environmental factors taken into consideration along with the standard GDP of a country. Green GDP is a term used generally for **expressing GDP after adjusting for environmental damage.** **Hence option (b) is the correct answer.**

- The standard GDP measurement has limitations on account of being indicators of economic growth and desirable standards of living. The normal GDP also does not have any way of knowing whether the level of income created in a country will be sustainable or not. To overcome this limitation the green GDP is sought after.
- The process of environmental accounting involves three steps viz. **Physical accounting; Monetary valuation; and integration with national Income/wealth Accounts.** Physical accounting determines the state of the resources, types, and extent (qualitative and quantitative) in spatial and temporal terms. Monetary valuation is done to determine its tangible and intangible components. Thereafter, the **net change in natural resources in monetary terms is integrated into the Gross Domestic Product** in order to reach the **value of Green GDP.**

Q 50. B • The Satavahana kingdom had three grades of feudatories. The highest grade was formed by the king who was called **Raja** and had the right to strike coins. The second grade was formed by the **mahabhoja** and the third grade by the **Senapati**. It seems that these feudatories and landed beneficiaries enjoyed some authority in their respective localities. **Hence statement 1 is not correct.**

- The Satavahana rulers were Brahmanas, and they represented the march of triumphant Brahmanism. From the very outset, **kings and queens performed such Vedic sacrifices as ashvamedha vajapeya** paying liberal sacrificial fees to the brahmanas. They also worshipped a large number of Vaishnava gods such as Krishna and Vasudeva. However, the Satavahana rulers promoted Buddhism by granting land to the monks. In their kingdom, the Mahayana form of Buddhism commanded a considerable following, especially amongst the artisan class. Nagarjunakonda and Amaravati in AP became important seats of Buddhist culture under the Satavahanas, and more so under their successors, the Ikshvakus. Similarly, Buddhism flourished in the Nasik and Junar areas in western Deccan in Maharashtra, where it seems to have been supported by traders. **Hence statement 2 is correct.**
- **The official language of the Satavahanas was Prakrit.** All their inscriptions were composed in this language and written in the Brahmi script, as was the case in Ashokan times. Some Satavahana kings may have composed Prakrit books. One Prakrit text called Gathasattasai, or the Gathasaptasati, is attributed to a Satavahana king called Hala. It consisted of 700 verses, all written in Prakrit, but it seems to have been finally re-touched much later, possibly after the sixth century.

Q 51. B • The Small Finance Bank (SFB) is a private financial institution intended to further the objective of financial inclusion by primarily undertake basic banking activities of acceptance of deposits and lending to un-served and underserved sections including small business units, small and marginal farmers, micro and small industries and unorganized sector entities, but without any restriction in the area of operations, unlike Regional Rural Banks or Local Area Banks. Small Finance Banks were created pursuant to the announcement in Union Budget 2014-2015.

- Small Finance Banks (SFBs) in India are required to maintain a minimum paid-up voting equity capital of ₹200 crore. This requirement applies to on-tap licensing, with exceptions for certain conversions from UCBs or NBFCs, and mandates a minimum capital adequacy ratio of 15%. **Hence statement 1 is not correct.**

- SFBs are required to open at least 25% of its branches in unbanked rural centres. **Hence statement 2 is correct.**
- Further, these banks are required to extend 75% of its ANBC to priority sector.

RBI's 2024 Guidelines for Converting SFBs into Universal Banks

- Eligible Applicants: **Only listed Small Finance Banks (SFBs) are eligible to apply for conversion** into a Universal Bank. **Hence statement 3 is correct.**
- Financial Requirements: Must have a minimum net worth of Rs 1,000 crore, scheduled bank status, and a profitable operational record for at least five years.
- Asset Quality Criteria: Must maintain gross NPAs below 3% and net NPAs below 1% consistently for the previous two years.

Q 52. D Directorate of Revenue Intelligence (DRI)

- It is the apex anti-smuggling agency of India, established in 1957.

Ministry

- **Functions under the Central Board of Indirect Taxes and Customs (CBIC), Ministry of Finance.**

Functions

- Enforces the provisions of the Customs Act, 1962 and more than fifty allied Acts, including: Arms Act, **Narcotic Drugs and Psychotropic Substances (NDPS) Act**, 1985 COFEPOSA, Wildlife (Protection) Act, Antiquities and Art Treasures Act, etc.

Responsibilities

- Detecting and preventing smuggling of contraband, including illegal drug trafficking.
- **Preventing and detecting illicit international trade in wildlife.**
- Combating commercial frauds related to international trade and evasion of customs duty.
- **The Directorate of Revenue Intelligence (DRI) is not a statutory body established by a specific Act of Parliament; rather, it is an attached office under the Central Board of Indirect Taxes and Customs (CBIC), Ministry of Finance, founded in 1957. It acts as India's apex anti-smuggling agency, enforcing the Customs Act of 1962. Hence, option (d) is the correct answer.**

Q 53. B • Article 293 of the Indian Constitution provides provisions related to borrowing by states.

- **Article 293(1) of the Indian Constitution provides that a state government can borrow within India against the security of its Consolidated Fund without the need for prior consent from the Central Government if they have no outstanding Central loans. Hence, statement 1 is correct.**
- **Article 293(3) explicitly states that if a state has an outstanding loan from the Central Government, it requires prior consent from the Centre before it can raise further loans. This provision ensures that the Central Government has oversight on states that are indebted to it, maintaining fiscal discipline. Hence, statement 2 is correct.**
- **Article 293(4) explicitly provides that the Centre may impose conditions when granting consent for state borrowing. The Centre controls the borrowing limits of states under the Fiscal Responsibility and Budget Management (FRBM) Act to maintain macroeconomic stability.**

- **The Indian Constitution does not explicitly prohibit states from raising loans from foreign sources.** Foreign borrowing falls under the Union List, and hence, only the Central Government has the power to raise loans from foreign sources (as per Article 246 and the Union List, Entry 37).
- However, **in April 2017, the Union Cabinet approved the policy guidelines to allow financially sound State Government entities to borrow directly from bilateral ODA** (Official Development Assistance) partners like JICA for the implementation of vital infrastructure projects. These guidelines facilitate the State Government entities to directly borrow from the external bilateral funding agencies, subject to the fulfilment of certain conditions, and all repayments of loans and interests to the funding agencies are to be directly remitted by the concerned borrower. **The concerned State Government has to furnish a guarantee for the loan. The Government of India provides a counter-guarantee for the loan.**
- Hence, statement 3 is not correct.
- Hence, option(b) is the correct answer.

- Q 54. A**
- Prepaid Payment Instruments (PPIs) are a convenient way to make purchases by storing funds in advance on an instrument, such as a mobile wallet, card, or voucher, which is then used to buy goods and services.
 - They include **closed systems for a single issuer, semi-closed systems for multiple authorised merchants, and open systems** (usually bank-issued) for broader use and cash withdrawal. Common examples are digital wallets like Paytm, gift cards, metro rail cards, and paper vouchers, offering users secure, budget-friendly, and flexible payment solutions.
 - **PPIs are instruments that facilitate the purchase of goods and services, the conduct of financial services, enable remittance facilities, etc., against the value stored therein. E.g. Mobile wallets, digital wallets, gift cards.**
 - **PPIs can be issued by banks and non-banks.** Classified under two types: small PPIs (issued after obtaining minimum details of the PPI holder) and Full KYC PPIs.
 - Debit cards are not themselves prepaid payment instruments (PPIs), but rather are often used to load money into a PPI. PPIs facilitate purchases against a pre-stored value, which can be loaded using a debit card, credit card, or by cash. Debit cards are open-system instruments linked to a bank account, allowing for payments and cash withdrawals but are not prepaid in nature.
 - Pre-paid Payment Instruments (PPIs) are defined in the RBI Guidelines issued under the Payment and Settlements Systems Act, 2007. These are regulated by the Reserve Bank of India. **Hence option (a) is the correct answer.**

- Q 55. C**
- Nitrate contamination in groundwater is a primary cause of methemoglobinemia, commonly known as "blue baby syndrome". In infants, nitrates are converted into nitrites, which bind to haemoglobin and reduce the blood's ability to carry oxygen. **Hence, statement 1 is correct.**
 - Agricultural runoff contains high levels of nitrogen and phosphorus from fertilisers. When these nutrients enter water bodies, they stimulate the rapid growth of algae, a process known as eutrophication or algal blooms. **Hence, statement 2 is correct.**
 - Nitrogen pollution originates from point sources, such as industrial discharge pipes and sewage treatment plants, as well as non-point sources (diffuse sources), such as agricultural runoff from fields and atmospheric deposition. **Hence, statement 3 is correct.**

- Q 56. A** • Behdienkhlam is a traditional festival of the Pnar or Jaintia tribe. **Hence statement 1 is correct**
- It celebrated after sowing is done seeking a good harvest and to drive away plague and diseases. On the other hand harvest festivals are those which are celebrated after the main harvest of a particular region. **Hence statement 2 is not correct.**
 - Young men symbolically drive away evil spirits by beating the roof of every house with bamboo poles. These tribesmen carry 'Rongs' or chariots and dance in muddy waters as others watch. Women do not participate in the dancing.
- Q 57. D** • **All the statements are correct.**
- **The republican system of government existed either in the Indus basin or the foothills of the Himalayas in eastern UP and Bihar.** The republics in the Indus basin may have been the remnants of the Vedic tribes, although some monarchies may have been followed by republics. In some instances in UP and Bihar, people were possibly inspired by the old ideals of tribal equality which did not give much prominence to the single raja.
 - **In the republics, real power lay in the hands of tribal oligarchies.** In the republics of Shakyas and Lichchavis, the ruling class belonged to the same clan and same varna.
 - In a tribal oligarchy, each raja was free to maintain his own little army under his Senapati, enabling each of them to compete with the other. **The Brahmanas exercised great influence in a monarchy, but they had no place in the early republics, nor did they recognize these states in their law-books.**
- Q 58. A** • The Praja Mandal Movements were directed against both feudalism and colonialism. People living in the princely states had to resist the autocratic rule of their feudal princes as well as the overarching control of the British administration. The central demand of these movements was the extension of democratic rights to the people, ensuring greater participation in governance and freedom from oppressive rule. **Hence, statement 1 is correct, and statement 2 is not correct.**
- The Congress initially maintained a policy of non-interference regarding the Praja Mandal movements in princely states during the 1920s to focus on British India, but this stance shifted significantly toward support by the late 1930s. The Haripura Session (1938) marked a policy change where Congress formally supported the struggle for democratic rights, though it often encouraged local leaders to operate under independent banners rather than directly under the Congress name. **Hence, statement 3 is not correct.**
 - The policies and programs of the Indian National Congress played a decisive role in shaping and strengthening the Praja Mandal Movement. The Gandhian methods of non-cooperation and civil disobedience, the Congress's role in organising the All India States People's Conference, and the declaration at the Tripuri Session to involve the states in the national struggle gave much-needed confidence and direction to the people's organisations. By joining hands with the People's Council, Congress made the princely state's struggles part of the larger national movement to end British rule and integrate the states into independent India.
- Q 59. D** • Central Vigilance Commission is one of the oldest autonomous Integrity Institutions of the world. The Central Vigilance Commission was set up by the Government in 1964 on the recommendations of the Committee on Prevention of Corruption, headed by Shri K. Santhanam, to advise and guide

Central Government agencies in the field of vigilance. **In 2003, the Parliament enacted a law conferring statutory status on the CVC. Hence statement 1 is not correct.**

- The CVC is not controlled by any Ministry/Department. It is an independent body which is only responsible for the Parliament. Its members are appointed by President on the recommendation by a selection panel consisting of PM, Leader of Opposition in LS and Minister of Home Affairs. President can remove chief Vigilance commissioner and other commissioners as per the provisions made in the CVC Act only. **Hence statement 2 is correct.**
- It is not an investigating agency. The CVC either gets the investigation done through the CBI or through chief vigilance officers (CVO) in government offices. The CVC has to present annually to the President a report on its performance. The President places this report before each House of Parliament. **Hence statement 3 is not correct**

Hence option (d) is the correct answer.

Q 60. D Engel's Law

- Engel's Law explains how household consumption patterns change with rising income. **It states that as income increases, the proportion of income spent on food declines**, even though absolute spending on food may still rise. This reflects the fact that food is a basic necessity and does not scale proportionately with income growth. **Hence, statement 2 is not correct.**

Understanding Income Elasticity of Demand for Food

Income elasticity of demand measures how much the demand for a good changes when income changes.

- **Food items generally show income-inelastic demand. Hence, statement 1 is not correct.**
- As income rises, people do not increase food consumption at the same rate as income.
- Instead, additional income shifts toward non-essential goods and services such as education, recreation, health, and lifestyle products.

Q 61. B • Recent Context: 30,000 live animals were seized under Operation Thunder 2025.

About Operation Thunder

- **It was coordinated by INTERPOL and the World Customs Organization (WCO), with the support of the International Consortium on Combating Wildlife and Forestry Crime (ICCWC).**
- It sought to intercept and seize illegally traded wildlife and forestry commodities across the global supply chain and identify, disrupt and dismantle criminal networks involved in these types of environmental crime.
- **Hence, option (b) is the correct answer.**

Q 62. D • Particularly vulnerable Tribal Groups (PVTGs) are a special category of tribal groups identified by the government as being more vulnerable than other tribal communities.

The Government of India, based on the recommendations of the Debar Commission, has established four main criteria to identify PVTGs –

- **Pre-agricultural Level of Technology** - This refers to the use of primitive tools for subsistence-level agricultural practice.
- **Stagnated or declining population** – PVTGs tend to have very low or negative population growth, which indicates their vulnerable demographic condition.
- **Extremely low literacy** – Low literacy level, particularly among adults, is a major criterion for identifying these groups.
- **Subsistence-level economy** - Many PVTGs practice a subsistence economy with limited market integration making them economically vulnerable.
- The Government of India has identified 52 PVTGs in 1975, which has increased to 75 in 1993.
- The GOI has launched 'The Scheme for Development of PVTGs' to adopt a holistic approach to the socio-economic development of PVTGs and gives state government flexibility in planning initiatives that are geared towards the specific socio-cultural imperatives of the specific group at hand.
- **Hence option (d) is the correct answer.**

- Q 63. A**
- Ghiasuddin Balban was last of the three strongest Sultans produced by Slave dynasty. In spite of the initial difficulties, he succeeded in restoring law and order; eliminating the conspirators; and safeguarding and keeping the Sultanate intact. He also propounded a new 12 theory of Kingship chiefly as an instrument of protecting his position as Sultan and to enhance the prestige of his office. His weak successors who started a new dynasty lost to the Khiljis.
 - Balban (r. 1266–1287) believed in absolute monarchy and the divine right of kings to counter the growing power of the Turkish nobility (the 'Forty'). He took the title Zil-i-Ilahi (Shadow of God) and introduced Persian court customs like Sijda (prostration) and Paibos (kissing the feet) to enhance the dignity and authority of the Sultan. **Hence, statement 1 is correct.**
 - Balban is known for his strict "blood and iron" policy to suppress internal revolts and maintain law and order. He was ruthless, often executing rebels, robbers, and their families, and even killing innocent civilians, such as in the case of the Mewatis and in his campaign against Tughril Khan in Bengal. **Hence, statement 2 is not correct.**

Q 64. D The Variable Repo Rate (VRR) and Fixed Repo Rate:

- The Variable Repo Rate (VRR) and Fixed Repo Rate are mechanisms employed by central banks, such as the Reserve Bank of India (RBI), to manage liquidity in the financial system. Understanding their differences is crucial for comprehending monetary policy operations.

Tenure of Borrowing:

- **Fixed Repo Rate:** Primarily used for overnight borrowing under the Liquidity Adjustment Facility (LAF). Banks borrow funds from the RBI at a pre-determined fixed rate to meet short-term liquidity needs.
- **Variable Repo Rate (VRR):** Utilized for longer-term liquidity adjustments, typically through auctions with tenures extending beyond overnight, such as 7, 14, or 28 days. This approach allows the RBI to manage liquidity over extended periods.
- **Hence statement 1 is not correct.**

Collateral Requirements:

- **Both Fixed Repo Rate and VRR:** Require banks to pledge government securities as collateral. This collateralization is a standard practice in repurchase agreements to mitigate credit risk. **Hence statement 2 is not correct.**

Interest Rate Determination:

- **Fixed Repo Rate:** The interest rate is pre-determined by the RBI. Banks borrow funds at this fixed rate, which remains constant during the borrowing period.
- **Variable Repo Rate (VRR):** The interest rate is determined through a market-driven auction process. Banks submit bids at various rates, and the RBI allocates funds based on these bids, leading to a cut-off rate influenced by current market liquidity conditions.
- **Hence statement 3 is not correct.**

Q 65. C Odissi closely follows the tenets laid down by the Natya Shastra. Facial expressions, hand gestures and body movements are used to suggest a certain feeling, an emotion or one of the nine rasas. **The techniques of movement are built around the two basic postures of the Chowk and the Tribhanga.** The chowk is a position imitating a square - a very masculine stance with the weight of the body equally balanced. The tribhanga is a very feminine stance where the body is deflected at the neck, torso and the knees. **Hence, statement 1 is correct.**

For centuries *maharis* were the chief repositories of this dance. The *maharis*, who were originally temple dancers came to be employed in royal courts which resulted in the degeneration of the art form. Around this time, a class of boys called *gotipuas* were trained in the art, they danced in the temples and also for general entertainment. Many of today's gurus of this style belong to the *gotipua* tradition. In each performance, even a modern Odissi dancer still reaffirms **the faith of the devadasis or maharis where they sought liberation or moksha through the medium of dance.** **Hence, statement 2 is correct.**

- Q 66. A**
- Cost-push inflation is inflation caused by **an increase in prices of inputs like labor, raw material, etc.** The increased price of the factors of production leads to a **decreased supply of these goods.** **While the demand remains constant,** the prices of commodities increase causing a rise in the overall price level. This is in essence cost-push inflation. In this case, the overall price level increases due to higher costs of production which reflects in terms of increased prices of goods and commodities which majorly use these inputs. This is inflation triggered from the supply-side i.e. because of less supply. The opposite effect of this is called demand-pull inflation where higher demand triggers inflation. **Hence statement 1 is correct**
 - Apart from the rise in prices of inputs, there could be other factors leading to supply-side inflation such as **natural disasters, global pandemics such as COVID-19 or depletion of natural resources, monopoly, government regulation or taxation, change in exchange rates,** etc. Generally, cost-push inflation may occur in the case of an inelastic demand curve where the demand cannot be easily adjusted according to rising prices.
 - Recently the global pandemic of COVID-19 acted as a double-sided sword wherein India witnessed a cost-push as well as Demand-Pull inflation. The mass exodus of migrant laborers from industrial belts led to a decreased supply of manpower and thus leading to reduced production.
 - **Demand-pull inflation (and not cost-push inflation) is an indicator of a growing economy** as when consumers feel confident, they spend more and take on more debt. This leads to a steady increase in demand, which means higher prices. **Hence statement 2 is not correct.**

- Q 67. B** • E-waste, or electronic waste, refers to discarded electrical and electronic devices like phones, computers, and appliances, containing valuable materials but also hazardous substances (lead, mercury) that harm health and the environment if improperly handled. It's the world's fastest-growing waste stream, driven by rapid tech cycles and consumption, with only a fraction formally recycled, creating significant risks, especially in developing nations. Proper recycling recovers resources (gold, copper) and minimises pollution, but informal methods often release toxins, impacting communities and soil/water.

Based on standard environmental data and regulations:

- Mercury is a highly toxic neurotoxin found in fluorescent lamps, switches, relays, and flat-screen monitors. **Hence, point 1 is correct.**
- Cadmium is commonly used in rechargeable batteries (NiCd), resistors, and CRT screen coatings. It is a known carcinogen that causes kidney and lung damage. **Hence, point 2 is correct.**
- Beryllium is found in computer motherboards and connectors. Chronic exposure to its dust or fumes can cause "berylliosis," a serious lung disease, and it is classified as a human carcinogen. **Hence, point 3 is correct.**
- While Platinum is found in electronic waste (particularly in printed circuit boards and hard disks), it is classified as a precious metal rather than a "harmful material" in the context of toxic e-waste pollutants. It is a valuable resource targeted for recovery and recycling rather than a primary hazardous waste concern like the others listed. **Hence, point 4 is not correct.**

- Q 68. D** • **The Danish East India Company was established in 1616 and, in 1620, they founded a factory at Tranquebar near Tanjore, on the eastern coast of India.**
- **Their principal settlement was at Serampore near Calcutta.**
 - The Danish factories, which were not important at any time, were sold to the British government in 1845.
 - **The Danes are better known for their missionary activities than for commerce.**
 - **Hence, option (d) is the correct answer.**

- Q 69. C** List of some important Books, magazines, newspapers edited/written by freedom fighters:

- Subhash Chandra Bose – Indian Struggle
- **Muhammad Ali – Comred, Hamdard**
- **Bhogaraju Pattabhi Sitaramayya – History Of Congress**
- Valentine Chirol – Indian Unrest
- **Chittaranjan Das – India For Indians**
- Bankim Chandra Chattopadhyay – Anand Math, Devi Chaudharani
- **Bhartendu Harishchandra – Bharat Durdasha**
- Shivanand – Divine Life
- Nayantra Sahgal – A Voice Of Freedom
- M.K. Gandhi (Mahatma Gandhi) – Young India, Harijan, Navjivan, Hindu Swarajya, My Experiments With Truth
- Bal Gangadhar Tilak – Kesri, The Maratha, Gita Rahasya, The Arctic Home in Vedas

- Jawahar Lal Nehru – Discovery of India, Glimpses of World History, Meri Kahani
- Maulana Abdul Kalam Aazad – Al Hilal, India Wins Freedom, Gubare Khatir
- Lala Lajpat Rai – Unhappy India, biographies of Mazzini, Garibaldi, Shivaji and Shrikrishna
- Dr. Rajendra Prasad – India Divided
- Lala Hardayal – Hints For Self Culture
- Surendra Nath Banarji – Bangali, A Nation In Making
- Veer Savarkar – War Of Indian Independence
- Ravindra Nath Tagore – Geetanjali (Nobel Prize Winner Book), Home and The World, Gora, Hungry Stones, Chandalika, Visarjan
- Madan Mohan Malviya – Abhyuday, Hindustan, Leader

Q 70. D Charter Act, 1853

- It was considered a vital constitutional landmark. **The Indian(central) legislative council was established.** Legislation, for the first time, was treated as a special function of the government, requiring special machinery and special process.
- Local representation was introduced in the Indian (Central) Legislative Council. The covenanted civil service was made available to Indians as well.
- The act separated the legislative and executive functions of the Governor General's council.
- It established a separate Governor-General's legislative council which came to be known as the Indian (Central) Legislative Council and provided for the addition of six new members called legislative councillors to the council.
- This legislative wing of the council functioned as a mini-Parliament, adopting the same procedures as the British Parliament. It established an open competition system for selection and recruitment of civil servants. The covenanted civil service was thus thrown open to the Indians also. Accordingly, the Macaulay Committee (the Committee on the Indian Civil Service) was appointed in 1854.
- It asked for the appointment of six new legislative councillors to the council. In other words, it constituted a distinct legislative body for the Governor-General, which became known as the Indian (Central) Legislative Council. The council's legislative branch operated as a mini-Parliament, using the same processes as the British Parliament. For the first time, legislation was considered a distinct government activity, needing specialised apparatus and procedures.

Hence, option (d) is the correct answer.

- Q 71. D**
- The names **THEJAS64, DHRUV64, and VIKRAM are related to Microprocessors.** These are key components of India's indigenous semiconductor ecosystem developed under the Microprocessor Development Programme (MDP) and the Digital India RISC-V (DIR-V) initiative.
 - DHRUV64: India's first fully indigenous 1.0 GHz, 64-bit dual-core microprocessor. Launched in December 2025, it was developed by C-DAC for use in 5G infrastructure, industrial automation, and strategic systems.
 - THEJAS64: A 64-bit single-core System-on-Chip (SoC) developed by C-DAC. It was the second chip fabricated under the DIR-V program and is primarily designed for industrial automation.

- VIKRAM: A microprocessor developed by ISRO and the Semiconductor Laboratory (SCL) specifically for space applications. It is engineered to withstand extreme space conditions and is used for navigation, guidance, and mission operations.
- **Hence, option (d) is the correct answer.**

- Q 72. A**
- The women belonged to diverse backgrounds and fought many battles to shape the Indian Constitution. However, none of these women were a part of the Drafting Committee of the Assembly.
 - The women members of the Assembly were Ammu Swaminathan, Dakshayani Velayudhan, Begum Aizaz Rasul, Durgabai Deshmukh, Hansa Jivraj Mehta, Kamla Chaudhry, Leela Roy, Malati Choudhury, Purnima Banerjee, Rajkumari Amrit Kaur, Renuka Ray, Sarojini Naidu, Sucheta Kriplani, Vijaya Lakshmi Pandit and Annie Mascarene.
 - They were lawyers, reformists, and freedom fighters and most of them were a part of the feminist movement. According to an analysis by PRS Legislative Research, most of the women that constituted the Assembly belonged to the upper caste, upper class and were literate. Out of the 15 women, only one was Muslim and another Dalit.
 - The zeal and contribution of the women members of the Assembly to discussions and debates on issues ranging from reservations to the Uniform Civil Code were unparalleled. Their contributions had a significant impact on several issues, such as gender equality, individual freedoms, decentralisation, and social justice.
 - Lakshmi Sahgal (Captain Lakshmi): Known for her role in the Indian National Army, she was not part of the assembly. **Hence, point 5 is not correct.**
 - **Hence, option (a) is the correct answer.**

- Q 73. B**
- In the wild, lion-tailed macaques are only native to the southwest of India. There, they live in the Western Ghats hills and mountains of South-western India, where they live in tropical rainforests and on elevated mountainsides. Its natural range is restricted to the rainforests of three states: Karnataka, Kerala, and Tamil Nadu. **Hence, statement 1 is correct.**
 - The Lion-tailed Macaque is not the smallest macaque species in India. While it is described as being "among the smaller macaques" or "one of the smallest macaque species in the world", the title of the smallest macaque species in India generally belongs to the Toque Macaque or smaller variations of the Rhesus Macaque, depending on specific body mass measurements. The Lion-tailed Macaque is a medium-sized primate, with adult males weighing up to 10 kg. **Hence, statement 2 is not correct.**
 - The species is classified as Endangered on the IUCN Red List of Threatened Species. It has maintained this status since 1996 due to severe habitat fragmentation and a declining wild population estimated at roughly 4,000 individuals. **Hence, statement 3 is correct.**

- Q 74. B**
- "Tiller of the Soil" is a famous 1938 tempera-on-paper painting by Indian artist Nandalal Bose, created as part of his renowned "Haripura Posters" series commissioned for the Indian National Congress. The artwork depicts a farmer (tiller) ploughing a field, showcasing rural life through folk-art style, earthy tones, and calligraphic, swift strokes, reflecting Gandhian values of dignity in labour. **Hence, Pair 1 is correctly matched.**

- Triumph of Labour is a famous 1959 bronze sculpture by Debi Prasad Roy Chowdhury located at Marina Beach, Chennai, and in New Delhi's National Gallery of Modern Art. It depicts four labourers struggling to move a rock, symbolising the dignity of manual labour and the working class. It was installed to commemorate India's first Labour Day. **Hence, Pair 2 is correctly matched.**
- Glow of Hope, alternately titled "Woman With the Lamp", is a painting by S.L. Haldankar. The work was painted in 1945–46 and is currently stored in the Sri Jayachamarajendra Art Gallery at the Jaganmohan Palace in Mysore, India. **Hence, Pair 3 is not correctly matched.**

- Q 75. A • The World Bank was established in July 1944 at the Bretton Woods Conference in New Hampshire, USA, primarily to rebuild Europe and Japan after World War II.** Originally known as the International Bank for Reconstruction and Development (IBRD), it began operations in 1946 to provide long-term capital for postwar reconstruction and development.
- The World Bank is like a cooperative, **made up of 189 member countries.** These member countries, or shareholders, are represented by a Board of Governors, who are the ultimate policymakers at the World Bank.
 - The five institutions are:
 - **International Bank for Reconstruction and Development (IBRD):** Provides loans and assistance to middle-income and creditworthy low-income countries.
 - **International Development Association (IDA):** Provides grants and low-interest loans to the poorest countries.
 - **International Finance Corporation (IFC):** Focuses on private sector development and investment.
 - **Multilateral Investment Guarantee Agency (MIGA):** Offers political risk insurance and credit enhancement.
 - **International Centre for Settlement of Investment Disputes (ICSID):** Provides arbitration for investment disputes.
 - Together, the IBRD and IDA form the World Bank. India is a member of all the above institutions except ICSID. **Hence option (a) is the correct answer.**

Q 76. D • Karez system

- The Karez system (also known as Qanat) is an ancient, gravity-based water harnessing technology consisting of a series of vertical shafts connected by a gently sloping underground tunnel. It is designed to:
- Tap Groundwater: The tunnel begins at a "mother well" located in an upland area or alluvial fan where the water table is high.
- Transport Water via Gravity: Because the tunnel's slope is gentler than the surface terrain, the water eventually emerges at the surface at a lower elevation.
- Prevent Evaporation: By moving water underground, the system minimizes loss to evaporation, making it highly effective in arid and semi-arid regions.
- In India, notable examples of this system (often introduced or expanded during the medieval period) are found in Bidar, Vijayapura (Bijapur), Aurangabad, and Burhanpur.

- Hence, option (d) is the correct answer.

Q 77. A • Torrefaction is a thermochemical pretreatment process that heats biomass to 200°C–320°C in an inert (low-oxygen) atmosphere to produce high-grade solid biofuel, often called bio-coal. It enhances biomass properties by increasing energy density, improving grindability, reducing moisture, and making the material hydrophobic. This process, often considered a form of "mild pyrolysis," transforms organic waste into a stable, combustible fuel that can replace coal in power generation

- Hence, option (a) is the correct answer.

Q 78. A • **The Supreme Court has the power to punish for contempt of itself under Article 129 of the Indian Constitution**, which states, "The Supreme Court shall be a court of record and shall have all the powers of such a court including the power to punish for contempt of itself."

- **Article 142 (2) explicitly grants the Supreme Court the power to investigate and punish any person for its contempt.** While this primarily focuses on contempt of the Supreme Court itself, it is understood in a broader context that the Supreme Court can exercise its authority to protect the judiciary's functioning at all levels.
- **In 1991, the Supreme Court ruled that it has the power to punish for contempt not only of itself but also of high courts, subordinate courts, and tribunals functioning in the entire country.**
- On the other hand, High Courts have been given special powers to punish contempt of subordinate courts, as per Section 10 of The Contempt of Courts Act of 1971.
- Hence, statement 1 is correct.
- **The Constitution of India does not explicitly define what constitutes contempt of court. This is defined in the Contempt of Courts Act, of 1971. Hence, statement 2 is not correct.**
- As per the Contempt of Courts Act 1971, contempt refers to the offense of showing disrespect to the dignity or authority of a court.
- **The act divides contempt into civil and criminal contempt:**
- **Civil contempt:** It is willful disobedience to any judgment, decree, direction, order, writ or other processes of a court or wilful breach of an undertaking given to the court.
- **Criminal contempt:** It is any publication that may result in Scandalising the court by lowering its authority, Interference in the due course of a judicial proceeding and obstruction in the administration of justice.
- Hence, statement 3 is not correct.

Q 79. D • **Tuberculosis (TB) is not transmitted by mosquitoes.** It is an airborne infectious disease caused by the bacterium *Mycobacterium tuberculosis*. It spreads from person to person through microscopic droplets released into the air when an infected individual coughs, sneezes, or speaks. **Hence, option (d) is the correct answer.**

- **Other options:**

- **(a) Malaria:** This is a life-threatening disease caused by *Plasmodium* parasites, which are transmitted to humans through the bites of infected female *Anopheles* mosquitoes.

- **(b) Dengue:** This viral infection is transmitted to humans through the bites of infected mosquitoes, primarily the *Aedes aegypti* species.
- **(c) Chikungunya:** This is a viral disease primarily spread to people through the bite of infected *Aedes aegypti* and *Aedes albopictus* mosquitoes.

Q 80. B Potholes: EROSIONAL

- These are circular, kettle-like depressions in the rocky bed of a river. They are formed by the abrasive action of pebbles and sand being whirled around in circular eddies (vortices). This "drilling" effect wears away the bedrock.

Natural Levees: DEPOSITIONAL

- Levees are raised embankments running parallel to the river. They form during floods when water spills over the banks and loses velocity, causing it to deposit its sediment load (primarily sand and silt) right at the edge of the channel.

Point Bars: DEPOSITIONAL

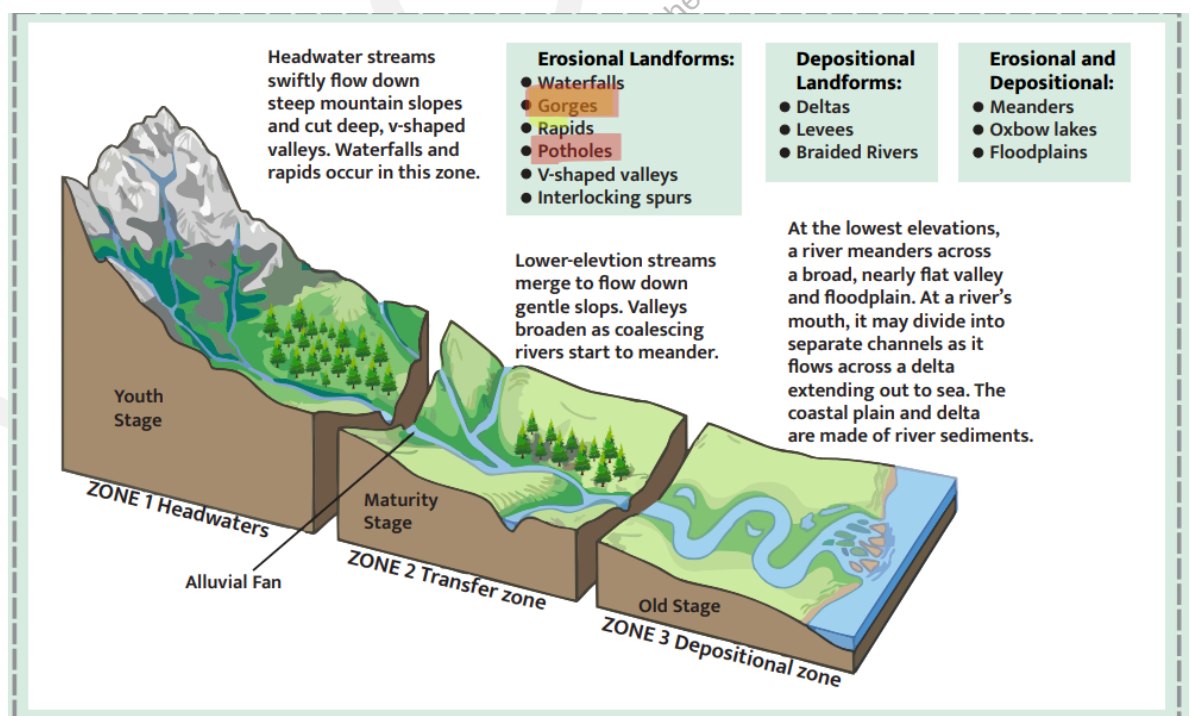
- Also known as meander bars, these are accumulations of sand and gravel found on the inside bend of a river meander. While the outside bend undergoes erosion, the inside bend has slower-moving water, which deposits sediment.

Gorges: EROSIONAL

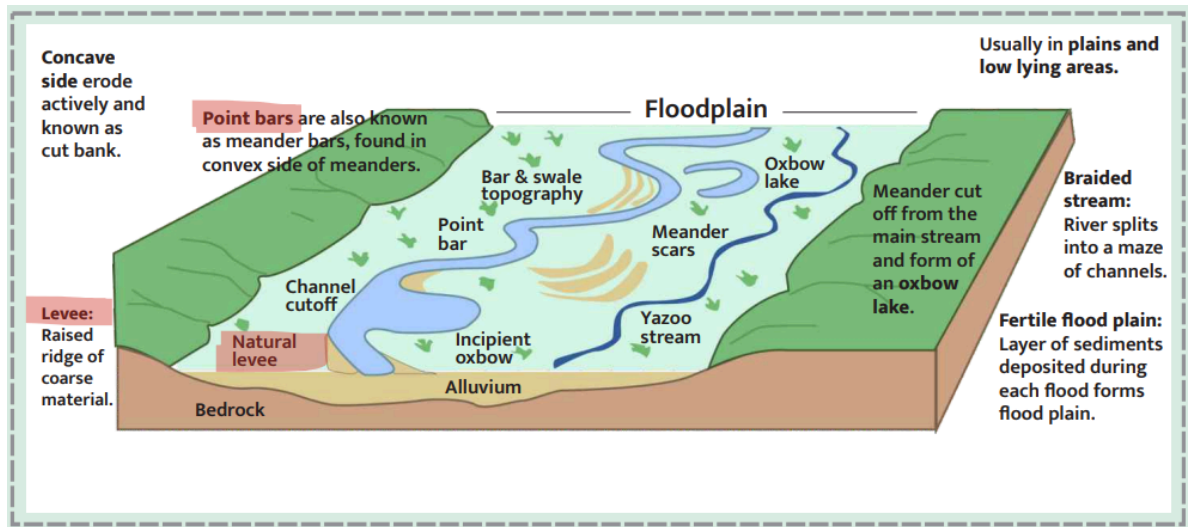
- A gorge is a deep, narrow valley with nearly vertical rocky walls. It is formed by the intense down-cutting (vertical erosion) of a river, typically in its youthful stage when it flows over hard rock or in regions of tectonic uplift.

- Hence, option (b) is the correct answer.

• **Erosional Landforms**



- **Depositional landforms**



- Q 81. C**
- The Ninth Schedule contains a list of central and state laws which cannot be challenged in courts. Currently, 284 such laws are shielded from judicial review. The Schedule became a part of the Constitution in 1951, when the document was amended for the first time. It was created by the new Article 31B, which along with 31A was brought in by the government to protect laws related to agrarian reform and for abolishing the Zamindari system. While A. 31A extends protection to 'classes' of laws, A. 31B shields specific laws or enactments. While most of the laws protected under the Schedule concern agriculture/land issues, the list includes other subjects, such as reservation. A Tamil Nadu law that provides 69 per cent reservation in the state is part of the Schedule.
 - The First Amendment added 13 laws to the Schedule. Subsequent amendments in 1955, 1964, 1971, 1974, 1975, 1976, 1984, 1990, 1994, and 1999 have taken the number of protected laws to 284.** Article 31B also has retrospective operation: meaning if laws are inserted in the Ninth Schedule after they are declared unconstitutional, they are considered to have been in the Schedule since their commencement, and thus valid. **Hence option (c) is the correct answer.**
 - Although Article 31B excludes judicial review, the apex court has said in the past that even laws under the Ninth Schedule would be open to scrutiny if they violated fundamental rights or the basic structure of the Constitution.

- Q 82. D**
- Mitochondria are the "powerhouses" of the cell and contain their own DNA (mtDNA). Mutations in this DNA can lead to devastating, often fatal, multi-systemic diseases. Since mitochondria are inherited exclusively from the mother, MRT allows a mother with faulty mitochondria to have a healthy biological child. **Hence, statement 1 is correct.**

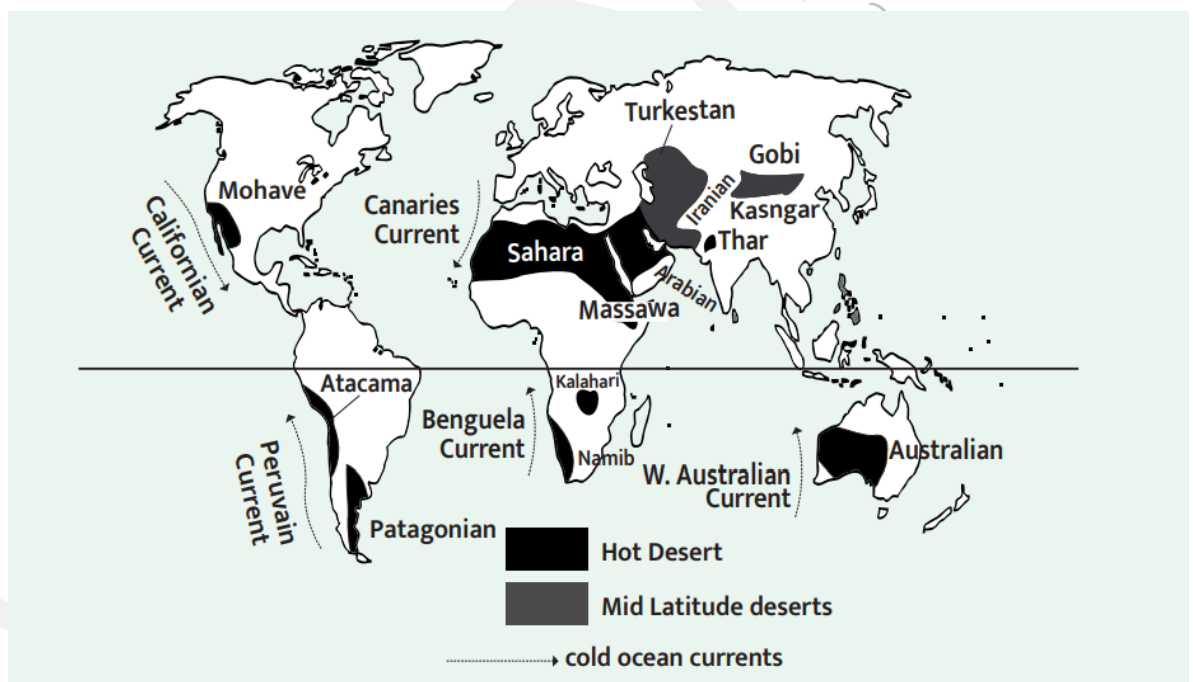
The primary techniques used are Spindle Transfer (before fertilization) or Pronuclear Transfer (after fertilization).

- The mother's egg contains "healthy" nuclear DNA but "unhealthy" mitochondria.
- A donor's egg contains "healthy" mitochondria but its nucleus is removed.
- The mother's nuclear material is inserted into the donor's egg. This egg now has the mother's genetic identity but the donor's healthy energy-producing machinery. **Hence, statement 2 is correct.**

A child born via MRT is genetically unique because they possess:

- Nuclear DNA (~99.8%): Inherited from the Mother and Father. This determines most physical traits (height, eye color, personality).
- Mitochondrial DNA (~0.2%): Inherited from the Female Donor. While it only contains 37 genes, it is a permanent part of the child's genetic makeup. Therefore, the child has genetic material from three people. **Hence, statement 3 is correct.**

- Q 83. B**
- The Patagonian Desert is located primarily in Argentina (South America). It is a cold winter desert and exists largely due to the rain-shadow effect created by the Andes Mountains to its west.
 - North American deserts include the Mojave, Sonoran, and Chihuahuan. **Hence, pair 1 is not correctly matched.**
 - The Kalahari is a vast semi-arid sandy savanna in Southern Africa, covering much of Botswana and parts of Namibia and South Africa. It is influenced by the cold Benguela current off the western coast. **Hence, pair 2 is correctly matched.**
 - The Great Victoria Desert is located in Australia (Western Australia and South Australia). It is the largest desert in Australia. **Hence, pair 3 is correctly matched.**
 - The Mojave Desert is located in the Southwestern United States (North America), primarily in California and Nevada. It is famous for Death Valley, the lowest and hottest point in North America.
 - South American deserts include the Atacama and the Patagonian. **Hence, pair 4 is not correctly matched.**



- Q 84. A**
- The Five Eyes Alliance is one of the key intelligence-sharing organizations with five English-speaking nations that include Australia, Canada, New Zealand, the United Kingdom, and the United States. **Hence statement 1 is not correct.**
 - Formed in World War II, its roots date back to the 1941 Atlantic Charter and were cemented in the UKUSA Agreement of 1946. .
 - Originally established for military intelligence gathering, the partnership has developed into a general framework for the exchange of signals intelligence (SIGINT) and for dealing with other international security issues. **Hence statement 2 is correct.**

- Recently, Australia made a decision to enlist non-citizen residents of the Five Eyes Alliance in its armed forces.

Q 85. C • The Central KYC Registry or CKYC is a centralized repository containing KYC records of customers with uniform KYC standards and interoperability. It is designed with the aim to reduce the burden of producing and verifying KYC documentation each time a customer establishes a new relationship with a financial entity. A CKYC ID is a unique ID assigned to you. It ensures that you do not have to submit your KYC documents multiple times for any financial transaction.

• **The Government of India authorized the Central Registry of Securitisation Asset Reconstruction and Security Interest of India (CERSAI)**, set up under the **Securitisation and Reconstruction of Financial Assets and Enforcement of Security Interest Act, 2002 to act as and to perform the functions of the Central KYC Records Registry. Hence, option (c) is the correct answer.**

Q 86. B • Higher Education Financing Agency (HEFA) is a joint venture of MoE Government of India and Canara Bank for financing creation of capital assets in premier educational institutions in India as part of rising 2022 HEFA's scope is greatly expanded to cover school education, educational institutes under Ministry of health etc. **Hence, statement 3 is correct.**

- HEFA is registered under Section 8 [Not-for-profit] under the Companies Act 2013 as a Union Govt company and as Non-deposit taking Systematically Important (NBFC-ND-SI) with RBI.
- HEFA incorporated on 31st May 2017, is a joint venture of MoE, GOI and Canara Bank with agreed equity participation in the ratio of 90.91% and 09.09% respectively
- Ours is a journey towards developing India's top-ranked institutions like IITs, IIITs, NITs, IISCs, and AIIMS into Globally top-ranking institutions through improvement in their academic and infrastructure quality. **Hence, statement 1 is not correct.**
- We are particularly interested in financing the building of educational infrastructure, R&D infrastructure and thereby enabling the institutions to reach top rankings globally. **Hence, statement 2 is correct.**

Q 87. A • The NASA-funded ATLAS (Asteroid Terrestrial-impact Last Alert System) survey telescope in Chile discovered an interstellar Comet named 3I/ATLAS.

- The Asteroid Terrestrial-impact Last Alert System (ATLAS) is a NASA-funded robotic survey and early warning system developed by the University of Hawaii. It consists of four to five telescopes in Hawaii, Chile, and South Africa that scan the entire sky every 24 hours to detect near-Earth objects (NEOs) days or weeks before potential impact.
- Although the ATLAS system is also used to identify dwarf planets, supernova explosions, and life exposure remanence from a star being absorbed by a supermassive black hole in a distant galaxy, the main purpose of this project is to search for large "killer asteroids" that may have a tremendous impact on life on earth.
- ATLAS will provide a warning time depending on the size of the asteroid — larger asteroids can be detected further from Earth.
- ATLAS will see a small (~20 meter) asteroid several days out, and a 100 meter asteroid several weeks out.

- The system is specially designed to detect objects that approach very close to Earth – closer than the distance to the Moon, about 240,000 miles or 384,000 kilometers away.
- **Hence, option (a) is the correct answer.**

Q 88. C The Bhakti movement (7th–17th century) was a transformative, pan-Indian, socio-religious revolution emphasising direct, emotional devotion (bhakti) to a personal God over rituals, caste distinctions, and orthodoxy. Originating with South Indian Alvars and Nayanars, it spread north, advocating for equality and using vernacular languages.

Features:

- The core of the movement was Bhakti (devotion), **emphasising an individual's intense, emotional, and direct bond with a personal god (such as Vishnu, Shiva, or Krishna).**
- Bhakti saints challenged the rigid social structures of the time. They **preached that devotion and love for God were more important than birth or status, making salvation accessible to all, including lower castes and women.**
- **Unlike the traditional elite-led practices conducted in Sanskrit, Bhakti poet-saints composed hymns and songs in local vernacular languages (like Tamil, Marathi, Hindi, and Bengali).** This allowed the common people to understand and participate in spiritual life.
- The Bhakti movement actually rejected the need for priestly intermediaries. **One of its main goals was to bypass complex rituals and the authority of the Brahmanical priesthood,** advocating instead for a direct personal connection with the divine.

Hence, option (c) is the correct answer.

- Q 89. D**
- As early as in **1957**, the Congress party had the bitter taste of defeat in Kerala. In the assembly elections held in March 1957, the Communist Party won the largest number of seats to the Kerala legislature. The governor invited **E. M. S. Namboodiripad, the leader of the Communist legislature party, to form the ministry.** For the first time in the world, **a Communist party government** had come to power through democratic elections.
 - In **1967** a peasant uprising took place in the **Naxalbari police station area** of Darjeeling hills district in West Bengal under the leadership of the local cadres of the **Communist Party of India (Marxist).** Beginning from the Naxalbari police station, the peasant movement spread to several states of India and came to be referred broadly as the Naxalite movement. In 1969, they broke off from the CPI (M) and a new party, Communist Party (Marxist-Leninist) (CPI-ML), was formed under the leadership of **Charu Majumdar.** It argued that democracy in India was a sham and decided to adopt a strategy of protracted guerrilla warfare in order to lead to a revolution.
 - In **1955, India nationalized Imperial Bank of India** with extensive banking facilities on a large scale, especially in rural and semi-urban areas. It renamed it as State Bank of India. **On 19th July 1969, a major process of nationalization was carried out and 14 major commercial banks in India** were nationalized. It was done during the Prime Ministership of **smt. Indira Gandhi.**
 - **India's first-ever coalition government** was formed at the national level under the **Prime Ministership of Morarji Desai,** which was also the first non-Congress national government, which existed from **24 March 1977 to 15 July 1979, headed by the Janata Party,** an amalgam of political parties opposed to Emergency imposed between 1975 and 1977. As the popularity of Janata Party dwindled, Morarji Desai had to resign and Charan Singh, a rival of Desai became the

fifth PM. However, due to lack of support, this coalition government did not complete its five-year term.

- **Hence option d is the correct answer.**

Q 90. A SODAR (Sonic Detection and Ranging) systems are remote sensing devices used primarily for measuring wind profiles, turbulence, and atmospheric boundary layer structure, typically at altitudes between 50m and 2000m.

Key Applications of SODAR Technology

- **Wind Energy Development:** SODAR is heavily used for measuring wind speed, direction, and shear for wind farm siting, performance optimisation, and resource assessment, often replacing or complementing traditional meteorological towers.
- **Meteorology and Atmospheric Studies:** Used for monitoring the atmospheric boundary layer (ABL), including measuring turbulence, thermal plumes, height of fog layer and inversion heights. **Hence, points 1 and 3 are correct.**
- **Aviation Safety:** Used in airport environments to monitor low-level wind shear, which is critical for safe takeoff and landing.
- **Environmental and Pollution Monitoring:** SODAR helps in understanding the dispersion of pollutants by mapping air movement and atmospheric stability, essential for studying air quality in urban areas. **Hence, point 4 is not correct.**
- **SODAR operates in the air.** Subsurface mapping typically uses seismic waves or ground-penetrating radar, not acoustic sounders designed for atmospheric profiling. **Hence, point 2 is not correct.**
- **While the technology is similar to Sonar (which is used underwater), SODAR is specifically the "in-air equivalent".** It is not designed to penetrate water or map the seabed. **Hence, point 5 is not correct.**

- Q 91. D**
- **An air mass is a large body of air, but its defining characteristic is its horizontal uniformity.** Within an air mass, temperature and moisture conditions are nearly the same at any given horizontal level (altitude).
 - **Vertical variation is normal; temperature usually decreases with height (lapse rate) within an air mass.** It is the horizontal consistency over hundreds of kilometers that makes it an "air mass." **Hence, statement 1 is not correct.**
 - **For an air mass to acquire uniform characteristics, it must stay over a surface for a long time.** This requires Source Regions that are vast, flat, and homogeneous (uniform in composition). Ideal source regions are large ocean plains (e.g., North Pacific) or vast level landmasses (e.g., Sahara Desert or Siberian Plains).
 - **High mountain ranges have irregular relief and cause turbulence/mixing, which prevents the air from becoming uniform.** **Hence, statement 2 is not correct.**
 - **When two different air masses (e.g., cold polar air and warm tropical air) meet, the boundary zone between them is called a Front.**
 - **The process of formation or intensification of a front is called Frontogenesis.**
 - **Frontolysis is the opposite; it refers to the dissipation or weakening of a front.** **Hence, statement 3 is not correct.**

- Q 92. B** • The Parliament Standing Committee on Personnel, Public Grievances, Law and Justice endorsed the establishment of regional benches of the Supreme Court, aligning with the recommendations of the 229th Law Commission Report, to address accessibility issues and decentralize judicial authority.
- Article 130 of the Indian Constitution: It says that the Supreme Court shall sit in Delhi or in such other place or places, as the Chief Justice of India (CJI) may, with the approval of the President, from time to time, appoint.

Hence option (b) is the correct answer.

- Q 93. A** Schedule I of the National Green Tribunal Act, 2010, lists seven key environmental legislations over which the National Green Tribunal (NGT) has jurisdiction for adjudicating cases, appeals, and compensation claims. These laws are essential for environmental protection, covering areas from water and air pollution to hazardous substances.

- The seven acts included in Schedule I are:
- The Water (Prevention and Control of Pollution) Act, 1974 **Hence, point 1 is correct.**
- The Water (Prevention and Control of Pollution) Cess Act, 1977
- The Forest (Conservation) Act, 1980
- The Air (Prevention and Control of Pollution) Act, 1981
- The Environment (Protection) Act, 1986
- The Public Liability Insurance Act, 1991 **Hence, point 2 is correct.**
- The Biological Diversity Act, 2002 **Hence, point 4 is correct.**
- Prevention of Cruelty to Animals Act, 1960, is not mentioned in Schedule I of the NGT Act 2010. The NGT's jurisdiction is limited to the seven specific environmental and conservation laws listed above. **Hence, point 3 is not correct.**

- Q 94. D** • Mangroves are indeed highly specialised, salt-tolerant (halophytic) trees and shrubs that thrive in tropical and subtropical intertidal zones, acting as essential, resilient buffers between land and sea. They are uniquely adapted to withstand high salinity, waterlogged (anoxic) soil, and constant tidal fluctuations. They exhibit the following feature:

- **Viviparity:** To survive in unstable mud and high salinity, many mangroves (like Rhizophora) exhibit vivipary, where seeds germinate while still attached to the parent tree.
- **Pneumatophores:** These are specialized "breathing roots" that grow upward out of the anaerobic, waterlogged soil to absorb oxygen directly from the air through pores called lenticels.
- **Salt glands on leaves:** Species like the black mangrove (Avicennia) use specialized salt glands on their leaf surfaces to actively excrete excess salt absorbed from the seawater.
- **Cutinized epidermis:** To minimize water loss (transpiration) in their physiologically dry (saline) environment, mangroves have a thick, waxy cutinized epidermis and often feature sunken stomata.
- **Hence, option (d) is the correct answer.**

- Q 95. C** • In Mexico and Central America, shifting cultivation is known as Milpa. Ladang is the local name used in Indonesia and Malaysia. **Hence, pair 1 is incorrectly matched.**

Philippines – Kaingin

- This is the traditional term used in the Philippines for the practice of clearing land for agriculture by cutting and burning. **Hence, pair 2 is correctly matched.**

Myanmar – Taungya

- Taungya is the Burmese term. Historically, it was adapted into the "Taungya System," a method where forest plantations were established in combination with food crops. **Hence, pair 3 is correctly matched.**

Brazil – Roca

- In Brazil, the practice of clearing a patch of land for temporary cultivation is locally termed Roca. **Hence, pair 4 is correctly matched.**

- Q 96. A • The Ricardian equivalence theorem (RET) plays an important role in macroeconomic theory.** The RET suggests that fiscal stimuli which are defined in terms of deficit-financed public spending hikes or tax cuts will lead to a crowding out of private consumption, thereby decreasing the effectiveness of fiscal policy in boosting economic activity. It states that **consumers are forward-looking** and will base their spending not only on their current income but also on their expected future income. **They will understand that borrowing today means higher taxes in the future.**

In simple words, if the government cuts taxes today and borrows money, people will save that extra money because they know taxes will increase later to repay the loan. **Hence option (a) is the correct answer.**

- It is called 'equivalence' because it argues that taxation and borrowing are equivalent means of financing expenditure. When the government increases spending by borrowing today, which will be repaid by taxes in the future, it will have the same impact on the economy as an increase in government expenditure that is financed by a tax increase today.
- **Okun's Law is based on the empirical research of Arthur Okun.** It describes the relationship between unemployment and growth rate in an economy. The law is still used by policymakers as a rule of thumb to estimate the relationship between growth rate and job creation.
- **Say's Law:** Often summarized as "supply creates its own demand," this 19th-century law suggests that the production of goods generates enough income to create the demand needed to purchase those goods.
- **Gresham's Law (Sir Thomas Gresham):** A monetary principle formulated in the 16th century stating that if two forms of commodity money (e.g., gold and silver coins) have the same face value but different intrinsic metal values, the "bad" (less valuable) money will drive the "good" (more valuable) money out of circulation.

- Q 97. A •** Like the Speaker, the Deputy Speaker is also elected by the Lok Sabha itself from amongst its members. He is elected after the election of the Speaker has taken place.
- The date of election of the Deputy Speaker is fixed by the Speaker. Whenever the office of the Deputy Speaker falls vacant, the Lok Sabha elects another member to fill the vacancy.
 - The Deputy Speaker performs the duties of the Speaker's office when it is vacant. He also acts as the Speaker when the latter is absent from the sitting of the House. In both cases, he assumes all the

powers of the Speaker. He also presides over the joint sitting of both Houses of Parliament, in case the Speaker is absent from such a sitting.

- It should be noted here that the Deputy Speaker is not subordinate to the Speaker. He is directly responsible to the House.
- **The Deputy Speaker has one special privilege, that is, whenever he is appointed as a member of a parliamentary committee, he automatically becomes its chairman. Hence, option (a) is correct.**
- Like the Speaker, the Deputy Speaker, while presiding over the House, cannot vote in the first instance; he can only exercise a casting vote in the case of a tie. Further, when a resolution for the removal of the Deputy Speaker is under consideration of the House, he cannot preside at the sitting of the House, though he may be present.

Q 98. B • Recent Context: In the Union Budget 2026-27, the Finance Minister recently proposed to raise the Securities Transaction Tax (STT) on both futures and options by up to 150% (0.02% to 0.03%).

- Securities Transaction Tax (STT) is a type of financial transaction tax levied in India on transactions done on the domestic stock exchanges.
- The rates of STT are prescribed by the Union Government through its Budget from time to time.
- In tax parlance, this is categorized as an indirect tax.

The STT came into effect from October 1, 2004, pursuant to the enactment of the Finance Act, 2004 and notification of Securities Transaction Tax Rules, 2004. The term 'Securities' is defined in the Securities Contracts (Regulation) Act and includes the following:

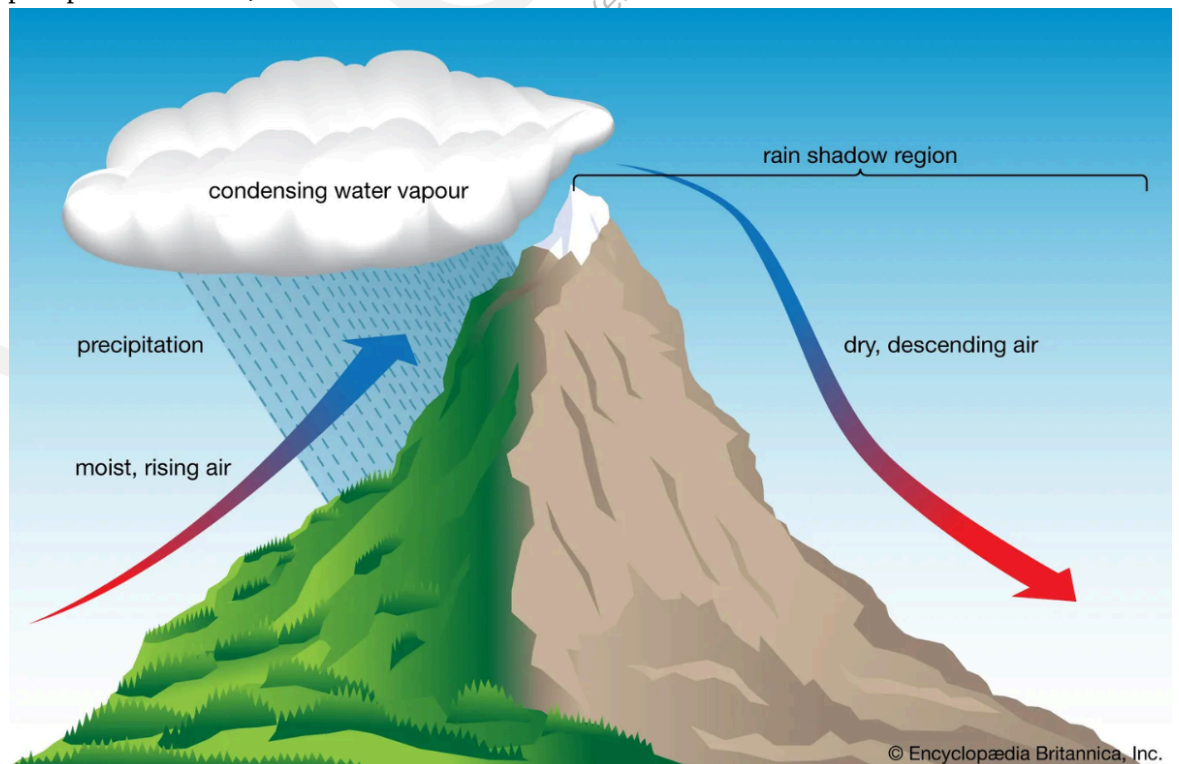
- Shares
- Bonds
- Debentures
- Equity Derivatives. **Hence, option 3 is correct.**
- Government securities of equity nature.
- Equity-oriented units of mutual fund.
- Securitized debt instruments.
- The rate of STT differs based on the type of security traded and whether the transaction is a purchase or a sale. These rates are decided by the central government. In India, STT is collected for the government of India by the stock exchanges.

STT is not applicable on:

- Gold ETFs, Gilt ETFs, International ETFs. **Hence, option 2 is not correct.**
- Commodity transactions (Commodities Transaction Tax is levied). **Hence, option 1 is not correct.**
- Currency transactions
- Transactions of the shares of listed companies on the floor of the stock exchange or otherwise such as a takeover, buyback, delisting offers etc also do not come under the STT framework.
- The off-market transactions of securities (which entails changes in ownership records at depositories) also does not attract STT.

- Q 99. B**
- The Extremist Phase (roughly 1905-1920) of the Indian National Movement marked a shift from moderate constitutionalism to assertive nationalism, led by figures like Bal Gangadhar Tilak, Lala Lajpat Rai, and Bipin Chandra Pal (the Lal-Bal-Pal trio).
 - Unlike the Moderates who relied on '3Ps' (Petition, Prayer, and Protest), the Extremists (Lal-Bal-Pal, Aurobindo Ghosh) believed in "passive resistance," which included direct action, Swadeshi (use of indigenous goods), boycotts, mass mobilization, and achieving Swaraj (self-rule) through methods like passive resistance, spurred by events like the Partition of Bengal in 1905. **Hence, statement 1 is correct.**
 - The Extremists utilized the Swadeshi movement to involve the masses, promote indigenous industry, and boycott British goods/institutions. **Hence, statement 2 is correct.**
 - In fact, the Extremists promoted national education as a key tool to foster nationalism, break free from British-controlled education, and promote indigenous culture. **Hence, statement 3 is not correct.**

- Q 100. C**
- The southern slopes of the Khasi Hills, specifically the villages of Mawsynram and Cherrapunji, consistently record the highest average annual rainfall not just in India, but globally. Mawsynram receives over 11,800 mm of rainfall annually. **Hence, statement I is correct.**
 - Location: The Khasi Hills are located at approximately 25°N latitude. This is well outside the equatorial zone (0° to 10°N/S). It is actually closer to the Tropic of Cancer.
 - Mechanism: The primary cause of the extreme rainfall is not equatorial convection. **Instead, it is Orographic Lift (relief rainfall).** The hills are shaped like a giant funnel. When the moisture-laden winds of the Bay of Bengal branch of the Southwest Monsoon strike these hills, they are forced to rise abruptly. As the air rises, it cools adiabatically, condenses, and results in heavy precipitation. **Hence, statement II is not correct.**



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